



OPEN Multi-strategy enhanced artificial rabbits optimization for prediction of grades in tourism service communication courses

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Predicting students' grades through their classroom behavior has been a longstanding concern in education. Recently, artificial intelligence has demonstrated remarkable potential in this area. In this study, the Artificial Rabbits Optimization Algorithm is selected to enhance the predictor's capabilities. This algorithm, a recently proposed and popular metaheuristic method, is known for its simple and straightforward structure. However, like other metaheuristic algorithms, it often falls into local optima, and as iterations increase, the convergence speed slows down, leading to reduced convergence accuracy. To address this issue, a Multi-Strategy Enhanced Artificial Rabbits Optimization Algorithm (MEARO) is introduced. MEARO first employs a Nonlinear Exploration and Exploitation Transition Factor (NL) to improve the balance between exploration and exploitation. Additionally, a Stochastic Centroid Backward Learning approach (SOBL) is applied to enhance both the quality and diversity of the population, ensuring a broader optimization of the search area and increasing the chances of locating the global optimum. Finally, a Dynamic Changing Step Length Development strategy is incorporated to enhance the randomness and development capability. The efficiency of MEARO is confirmed by comparing its performance with eight other sophisticated algorithms using the CEC2017 benchmark. The number of wins/Ties/losses in the three dimensions of cec2017 are (223/0/17), (221/0/19) and (230/0/10) respectively. Results indicate that MEARO outperforms these algorithms. Furthermore, MEARO is used to optimize two critical parameters of the Kernel Extreme Learning Machine (KELM), significantly improving its classification performance. Experimental results on the collected student performance dataset demonstrate that the KELM model optimized by MEARO surpasses other benchmark models across various metrics. Additionally, factors such as interest in the course, frequency of classroom discussion, and access to supplementary knowledge and information related to the course are identified as significant contributors to performance.

Keywords Artificial rabbits optimization, Intelligent education, Reinforced exploitation strategy, Stochastic centroid dynamic opposition-based learning, Kernel extreme learning machine

In contemporary society, the integration of artificial intelligence with education is a hot topic. However, this intersection covers a wide range, where employing AI technology to analyze key factors in education stands as one of the significant issues. We endeavor to identify key factors that enhance student performance in tourism service communication courses from an optimization perspective. Optimization is a natural process that seeks to find the best solution among multiple options. Generally, an optimization problem consists of three core elements: decision variables, an objective function, and constraints^{1–3}. The objective is to identify the best decisions for a group of parameters within given constraints to either maximize or minimize the objective function. To address such problems, there are two main algorithmic strategies. Deterministic algorithms are suitable for linear, continuous, differentiable, and convex problems, while stochastic algorithms, especially metaheuristic algorithms, are more adept at tackling complex real-world problems^{2,4,5}. Compared to deterministic algorithms, metaheuristic algorithms are favored for their simplicity and reliability. Metaheuristic techniques aim to find solutions that are close to the optimal, although they cannot provide strict guarantees of the global optimum, they often surpass exhaustive search methods in computational efficiency. These algorithms work by randomly

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adjusting initial candidate solutions and iteratively searching for the solution that meets the requirements or satisfies the stopping criteria^{6–8}.

As technology progresses, our society grapples with ever more intricate challenges in various fields like engineering, finance, and data analysis⁹. Conventional optimization techniques, including gradient descent and Newton's method, often fall short in tackling these complexities due to their reliance on specific problem frameworks and math models. To address this, metaheuristic algorithms come into play, offering a versatile approach to complex optimization¹⁰. These algorithms don't tie down to a problem's specific structure or mathematical representation; instead, they navigate the solution landscape using broad search strategies. Metaheuristic algorithms can be categorized into four main groups: Swarm Intelligence (SI), Evolutionary Algorithms (EA), Human-Inspired Algorithms (HA), and Physics-Based Algorithms (PhA). Among these, Swarm Intelligence stands out for its heuristic search approach, drawing inspiration from the collective behaviors observed in nature^{11,12}. These algorithms excel at solving complex problems by promoting interactions and information sharing within a group or swarm. The goal is to mimic the adaptive and cooperative behaviors seen in natural swarms, enabling Swarm Intelligence algorithms to seek optimal or near-optimal solutions within a problem's search range.

Although these algorithms have been successfully applied in other fields, their use in predicting grades for tourism service communication courses is relatively rare. In particular, research has yet to fully explore how swarm intelligence algorithms can be leveraged to model multidimensional student data, such as personal backgrounds, learning behaviors, and course participation. Furthermore, current studies lack a comprehensive approach to integrating educational content, student feedback, and external factors into prediction models through these optimization algorithms. The performance of students is influenced by a variety of complex factors, such as interest, engagement, learning habits, and attention, making it a very challenging issue. As a result, there is a clear research gap in how swarm intelligence algorithms could enhance the robustness, generalizability, and accuracy of grade prediction models in this context. Investigating the potential of these algorithms for grade prediction in tourism service communication courses could offer educators new tools to improve both teaching quality and student outcomes.

Literature review

In the realm of swarm intelligence algorithms, there's a rich array of approaches drawing inspiration from nature³. One notable example is the Particle Swarm Optimization (PSO) algorithm, pioneered by Eberhart and Kennedy, which takes cues from the flocking patterns of birds¹³. The Ant Colony Optimization algorithm (ACO) was proposed based on the foraging behavior of ants¹⁴. The Bat Algorithm (BA) was developed based on bats' echolocation for prey capture behavior¹⁵. The Grey Wolf Optimizer (GWO) draws its inspiration from the social hierarchy and hunting tactics observed in wolf packs¹⁶. The Harris Hawk Optimization algorithm (HHO) takes cues from the hunting methods of hawks going after rabbits¹⁷. Meanwhile, the Whale Optimization Algorithm (WOA) is modeled after the hunting behaviors of humpback whales, encompassing search, encircle, and spiral update stages¹⁸. The Grasshopper Optimization Algorithm (GOA) was proposed by mapping the short-range movement behavior of grasshopper nymphs to local exploration with short steps and the long-range movement behavior of adult grasshoppers to global exploration with long steps in a "steppe-tan synergy" manner¹⁹. The Marine Predator Algorithm (MPA) was inspired by marine predators' foraging behavior²⁰, is derived from the foraging and anti-predator behaviors observed in sparrows²¹. The Dung Beetle Optimization algorithm (DBO) was developed based on dung beetles' behaviors including rolling balls, dancing, foraging, stealing, and reproduction²². The Black Widow Optimization algorithm (BWO) takes inspiration from the distinctive breeding habits of black widow spiders²³. Finally, the Firefly Hunting Optimizer (FHO) was developed, taking inspiration from how fireflies hunt. It utilizes the distance between fireflies and their prey as a guiding factor for optimization. While these algorithms have common features, each also possesses unique traits that make them better suited for specific problem types.

Swarm intelligence optimization algorithms play a crucial role in addressing complex challenges. Firstly, they offer robust global search capabilities, steering clear of local optima pitfalls. Secondly, these algorithms are skilled at addressing high-Dimensional problems, nonlinear problems, and multimodal problems, which are often difficult for traditional mathematical methods to handle. Moreover, these algorithms thrive in parallel computing settings, harnessing computational power to expedite problem-solving processes^{24–26}. Their impact spans various fields. In engineering, they're commonly used for design optimization, power system scheduling²⁷, and drone path planning²⁸. In data science, they prove valuable for tasks like feature selection²⁹, clustering³⁰, network analysis, and fine-tuning machine learning model parameters³¹. Meanwhile, in finance, they find application in portfolio optimization and risk assessment.

The use of metaheuristic algorithms for optimizing machine learning models, especially in classification tasks, is gaining popularity. Because metaheuristic algorithms excel at fine-tuning the optimal parameters of machine learning models, these models often outperform their original versions. Among these, the optimization of neural network model parameters using swarm intelligence algorithms has become a trending area of interest. Many studies suggest that the performance of Kernel Extreme Learning Machine (KELM) classifiers is affected by two main parameters^{32–35}. These parameters are the kernel penalty parameter and the bandwidth. Thus, it's crucial to set these two parameters correctly before using the classifier in real-world applications. Grid search methods are commonly used to optimize the kernel penalty parameter and bandwidth. However, this method can be susceptible to local optima, and determining the parameter ranges can be a difficult task. To address these limitations, internal optimization methods have been incorporated within the framework. Optimization essentially aims to find certain optimal solutions or a set of unknown variable solutions. Several metaheuristic algorithms have been studied for this purpose, including Moth Flame Optimization (MFO)³⁶, Particle Swarm Optimization (PSO) and Harris Hawk Optimization (HHO)³⁷, Differential Evolution (DE)³⁸, Multi-Swarm

Meta-Heuristics Algorithm (MS-AFS)³⁹, Grey Wolf Optimization (GWO)¹⁶, Genetic Algorithm (GA)⁴⁰, Bacterial Foraging Optimization (BFO)⁴¹, Diversity-Oriented Social Network Search (DOSNS)⁴², Grasshopper Optimization Algorithm (GOA)⁴³, Harris Hawk Optimization (HHO)¹⁷, and Artificial Bee Colony (ABC)⁴⁴, improved firefly algorithm (GSFA)⁴⁵, Reptile Search Algorithm (RSA)⁴⁶. These algorithms strive to find the best parameters for the KELM model and are better equipped to achieve global optimal solutions than deterministic methods. Yet, these previously suggested optimizers come with their own set of parameters. The arbitrary setting of these parameters can impact their performance. In particular, the randomness of coefficient vectors in GWO can result in premature convergence when identifying the optimal parameters for KELM, even though these vectors are intended to enhance exploration^{34,47,48}. This indicates that there are still potential new optimizers that might be more effective for specific challenges. Moreover, as summarized by the “No Free Lunch” (NFL) theorem, there is no one-size-fits-all best method for handling all optimization cases⁴⁹. This fact has spurred the development of various different approaches, each aiming for superior optimization capabilities.

Moreover, the No Free Lunch (NFL)⁴⁹ theorem logically demonstrates that no algorithm performs optimally across all optimization problems. This has motivated many researchers to continually refine existing algorithms to tackle various optimization challenges. For example⁵⁰, provides a comprehensive review of metaheuristic algorithms and modifications applied to nature-inspired algorithms (NIAs) for feature selection (FS) problems. Moutaz Alazab et al. improved the Moth-Flame Optimization (MFO) algorithm by incorporating new operators outside of the embedded spiral operator to balance exploration and exploitation, addressing local minima issues⁵¹. In⁵², a binary variant of MFO, the Island-Based MFO (IsBMFO) model, was proposed to enhance optimization efficiency. Ruba Abu Khurma⁵³ introduced a feature selection approach based on a Levy Flight synchronized MFO for medical diagnosis. Additionally⁵⁴, proposed a biomedical Snake Optimizer, guided by a logarithmic survival global search, for optimizing feature selection in disease recognition applications. Abdelazim et al.⁵⁵ developed an improved Binary Whale Optimization (mBWO) algorithm that integrates elite evolutionary strategies, randomized control factors, and transitional factors to manage exploration and exploitation. Abu Khurma Abu Khurma developed a binary version of the Randomized Island Metaheuristic Evolution (RIME) algorithm, called mRIME, specifically for feature selection. These algorithms have been adapted to address the unique characteristics of their respective problems, achieving enhanced performance for each application⁵⁶.

The Artificial Rabbit Optimization algorithm (ARO)⁵⁷ is a recently introduced and popular metaheuristic algorithm known for its simple and intuitive structure, and for its fast convergence speed and convergence accuracy. However, similar to other metaheuristic algorithms, ARO often falls into local optima, and as the iterations progress, its convergence speed slows down, leading to reduced convergence accuracy. Thus, this study seeks to create an enhanced ARO for optimizing the Kernel Extreme Learning Machine (KELM) model for classification. We then plan to apply this ARO-KELM classification model to forecast students' exam scores. In the enhanced ARO, a nonlinear exploration and exploitation transition factor (NL) is employed to enhance the balance between exploration and exploitation in ARO; the Stochastic Optimal Backward Learning (SOBL) mechanism^{58,59} is adopted to improve the caliber and variety of the populace, ensuring that the algorithm can comprehensively optimize the search space and boost the chances of discovering the worldwide best solution; and a dynamic changing step size development strategy is employed to enhance the randomness and developmental capability of ARO. This enhanced version is known as Modified Enhanced ARO (MEARO). Subsequently, the MEARO algorithm is employed to fine-tune two crucial parameters of KELM, boosting the classification effectiveness of the KELM model. To evaluate the efficacy of the suggested approach, comprehensive assessments are conducted on CEC2017 benchmark functions and exam score forecast scenarios. Based on the assessment findings, we observe that the proposed MEARO outperforms the standard ARO and other renowned algorithms. Concurrently, the suggested approach can yield a more fitting KELM model and achieve superior prediction accuracy compared to several alternative models.

The main contributions of this study are as follows:

1. A new version of ARO, named Multi-strategy Enhanced ARO (MEARO), is proposed by integrating ARO with NL, SOBL, and an enhanced large development strategy.
2. The effectiveness of the proposed MEARO is validated through 30 experiments conducted on the CEC2017 optimization tasks.
3. The impacts of the three integrated mechanisms on the optimizer's performance are analyzed, and the performance of the proposed MEARO is compared with several other optimizers in solving CEC2017 test functions.
4. The recommended optimizer is effectively employed to create a grades prediction approach, MEARO-KELM, and applied to grades prediction. Simultaneously, within the collected data, interest in the course, frequency of classroom discussion, and access to extra knowledge and information related to the course emerged as the three most crucial factors influencing academic outcomes.

The structure of the paper unfolds as follows: Sect. 2 offers a concise overview of artificial rabbits optimization. Section 3 outlines the suggested approach. Section 4 showcases the related experiments and in-depth analyses. Section 5 employs the proposed MEARO for classification forecasting. Lastly, Sect. 6 encapsulates the study's conclusions and explores future directions.

Artificial rabbits optimization

Artificial rabbits optimization inspiration

The Artificial Rabbits Optimization (ARO) algorithm draws inspiration from the survival tactics of rabbits, which are encapsulated in its two primary search phases: exploration and exploitation. In the exploration phase, much like rabbits venturing far from their burrows, the algorithm casts a wide net across the solution space. It

uses a broad perspective, akin to a rabbit's expansive field of vision, to extensively explore potential solutions. Conversely, the exploitation phase reflects the evasion tactics of rabbits. When rabbits detect threats, they construct several burrows and randomly select one as their hideout. This unpredictability confounds predators. Similarly, the ARO algorithm diversifies its search by creating multiple solution paths, allowing it to adapt and respond to the evolving landscape of optimization challenges. Moreover, rabbits adjust their posture during these phases for swift and agile movement, increasing their odds of survival. In the ARO, these postural adjustments are symbolically represented through energy level adjustments. By modulating energy levels, the algorithm dynamically shifts between exploration and exploitation strategies, maintaining a balance⁶⁰. In essence, the ARO algorithm's design captures the essence of rabbits' adaptive survival behaviors, offering a flexible and efficient approach to optimization problems while maintaining a harmonious balance between exploration and exploitation.

Artificial rabbits optimization mathematical model

Initialization phase

Similar to many optimization algorithms that depend on a population of candidate solutions evolving through random processes, the ARO algorithm begins by producing a set of potential solutions randomly across the search area. This initial randomness sets the stage for the stages of the algorithm focused on exploring and exploiting opportunities, allowing it to navigate the solution landscape in search of achieving the best or nearly best solutions.

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_N \end{bmatrix}_{N \times dim} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,j} & \cdots & x_{1,dim} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i,1} & \cdots & x_{i,j} & \cdots & x_{i,dim} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,j} & \cdots & x_{N,dim} \end{bmatrix}_{N \times dim} \quad (1)$$

Let's use X to indicate the rabbit population size, X_i to show where the i^{th} rabbit is located, and $x_{i,j}$ to denote the position of the i^{th} rabbit in the j^{th} Dimension. N stands for the total number of rabbits in the area, while Dim specifies the Dimension of the rabbit locations.

The rabbits' starting locations are set randomly using this mathematical formula:

$$x_{i,j} = (upper - lower) \times rand + lower \quad (2)$$

in this context, $x_{i,j}$ stands for the starting value of the j^{th} decision factor for the i^{th} potential solution. The limits for the j^{th} decision factor of the i^{th} rabbit is denoted by $lower$ and $upper$. $rand$ is a random number uniformly spread between 0 and 1.

Detour searching (exploration)

In the ARO algorithm, the foraging detour behavior means that each search entity usually picks a random position from another entity in the group to update its own location. This action comes with controlled random adjustments. The research presents a mathematical framework for this detour behavior in rabbits, giving a quantitative explanation of their foraging habits. Here's the model:

$$X_i(t+1) = X_r(t) + R \times (X_i(t) - X_r(t)) + round(0.5 \times (0.05 + r_1)) \quad (3)$$

$$R = L \times c \quad (4)$$

$$L = \left(e - e^{\left(\frac{t-1}{T-1}\right)^2} \right) \times \sin(2 \times \pi \times r_2) \quad (5)$$

$$c(k) = \begin{cases} 1, & \text{if } k = g(l) \\ 0, & \text{else} \end{cases} \quad k = 1, 2, \dots, dim \text{ and } l = 1, 2, \dots, \lceil r_3 \times dim \rceil \quad (6)$$

$$g = randperm(d) \quad (7)$$

$$n_1 = N(0,1) \quad (8)$$

here, $X_i(t+1)$ stands for the i^{th} rabbit's potential location at the $(t+1)^{th}$ step, $X_i(t)$ shows where the i^{th} rabbit is currently at the t^{th} step, and $X_r(t)$ is the position of a rabbit chosen randomly from the N rabbits, excluding the one we're looking at. T is the highest iteration count, $\lceil \times \rceil$ is the ceiling function rounding up to the closest integer, and $round$ rounds a number to the nearest whole number. $randperm(d)$ produces a random integer between $(1, dim)$. Additionally, r_1, r_2 and r_3 random numbers between $(0,1)$. L represents how long they keep moving during detour foraging, and n_1 follows a standard normal distribution.

Random hiding (exploitation)

To evade predators, rabbits frequently use a strategy of multiple burrows for shelter. In the ARO algorithm, during each cycle, rabbits create Dim burrows across each Dimension of the search area and pick one at random to hide in. This method lowers their chances of getting caught. The location of the j^{th} burrow for the i^{th} rabbit is set by this formula:

$$X_{i,j}(t) = X_i(t) + H \times g \times X_i(t) \quad (9)$$

$$H = \frac{T-t+1}{T} \times n_2 \quad (10)$$

$$n_2 = N(0,1) \quad (11)$$

$$g(k) = \begin{cases} 1, & \text{if } k = g(l) \\ 0, & \text{else} \end{cases} \quad k = 1, 2, \dots, \dim \quad (12)$$

$$X_{i,j}(t+1) = X_i(t) + R \times (r_4 \times B_{i,r}(t) - X_i(t)) \quad (13)$$

$$g_r(k) = \begin{cases} 1, & \text{if } k = \lceil r_5 \times \dim \rceil \\ 0, & \text{else} \end{cases} \quad k = 1, 2, \dots, \dim \quad (14)$$

$$B_{i,r}(t) = X_i(t) + H \times g_r \times X_i(t) \quad (15)$$

where, $B_{i,r}(t)$ represents the randomly selected burrow for the i^{th} rabbit at iteration t . r_4 and r_5 are random values between (0,1). Once the detour searching or unplanned concealing tactic is implemented, the rabbit's position update is described as:

$$X_i(t+1) = \begin{cases} X_i(t), & \text{if } X_i(t) \leq X_i(t+1) \\ X_i(t+1), & \text{if } X_i(t) > X_i(t+1) \end{cases} \quad k = 1, 2, \dots, \dim \quad (16)$$

Energy reduction (shift from exploration to exploitation)

The ARO incorporates an energy factor to mimic the shift from exploration to exploitation. The formula for this energy factor is given by:

$$A(t) = 4 \times \left(1 - \frac{t}{T}\right) \times \ln \frac{1}{r_6} \quad (17)$$

where, r_6 is a random value between (0,1). If the energy factor $A(t) > 1$, the rabbit conducts foraging detour to explore in the algorithm. When $A(t) \leq 1$, the rabbit engages in random hiding for exploitation.

In summary, the diagram illustrating the ARO algorithm is depicted in Fig. 1.

Multi-strategy enhanced artificial rabbits optimization (MEARO)

Motivation

ARO exhibits superior optimization capabilities compared to many previously proposed metaheuristic algorithms. However, upon analyzing the optimization performance of ARO, it was found that its convergence speed is relatively slow, and like many other heuristic algorithms, it tends to become trapped in local peaks, making it difficult to find the global optimal solution. To overcome these shortcomings, we propose a Multi-Strategy Enhanced Artificial Rabbits Optimization (MEARO). This algorithm integrates multiple strategies, primarily including: Nonlinear exploration and over-exploitation factor, Stochastic centroid dynamic opposition-based learning strategy, and Reinforced exploitation strategy.

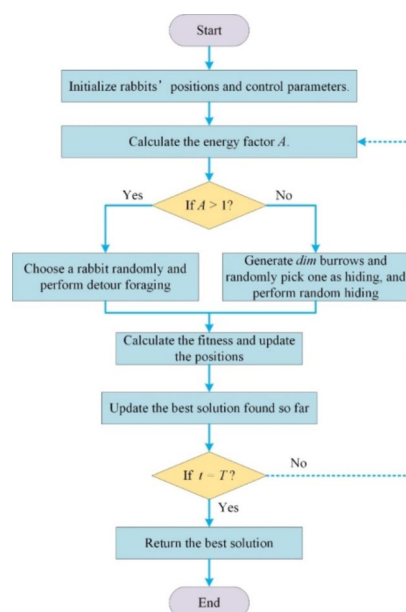


Fig. 1. Diagram of ARO algorithm.

Nonlinear exploration and over exploitation factor

To address the imbalance between exploration and exploitation in ARO, we use a Nonlinear Exploration and Over-exploitation Transition Factor to enhance the balance between exploration and exploitation in ARO. Transitioning from the exploration phase to the exploitation phase using a nonlinear factor, as opposed to a linear factor, allows the model to grasp intricate connections and patterns in the data. This enables the model to better adapt to intricate data structures, thereby improving its performance and accuracy. Additionally, it allows the model to be more flexible when facing new data, as it can adapt to different data distributions and patterns. This helps the model better cope with changes and uncertainties during the exploitation phase. Therefore, the improved nonlinear transition factor equation is as follows:

$$A(t) = 4 \times \left(1 - \frac{t}{T}\right)^{2 \times \frac{t}{T}} \times \ln \frac{1}{r_6} \quad (18)$$

The comparison between the improved nonlinear transition factor and the original linear transition factor is shown in Fig. 2.

Stochastic centroid dynamic opposition-based learning strategy

In problem-solving, one usually starts from a random or zero point and gradually moves towards the optimal solution. This applies to parameters like weights in neural networks, parameters in swarm intelligence algorithms, kernel parameters in support vector machines, and so on. If the initial random values are near the best solution, the problem can be solved quickly; however, if the initial values are distant from the best solution, it could take a considerable amount of time. Obtaining good initial values without prior knowledge can be challenging. Logically, the solving process can explore multiple directions. Introducing the reverse of the optimal solution as feasible solutions expands the search space, enhancing optimization efficiency, which is the core of reverse learning. However, pure reverse learning may encounter the problem of local optima. Therefore, various variants of reverse learning have emerged, such as random reverse learning, population incremental learning, lens imaging reverse learning, and stochastic centroid reverse learning, among others⁵⁸.

To address the sluggish convergence rate and reduced precision issues of ARO, this study integrates the stochastic centroid dynamic opposition-based learning (SOBL) mechanism into the ARO algorithm. This approach helps to improve both the quality and variety of the population. It ensures that the algorithm can comprehensively optimize the search space, thereby increasing the probability of finding the global optimal solution. The steps of the SOBL are as follows:

Firstly, after each iteration, produce a random whole number K from the interval $[1, N]$, where N is the population size. Select the leading K individuals with the highest fitness from the current population, denoted as $X_1, X_2, \dots, X_{K-1}, X_K$. Secondly, compute the centroid of these K individuals, as shown in Eq. (19):

$$C = \frac{\sum_{i=1}^K K_i}{K} \quad (19)$$

Next, for each individual at the centroid position, generate its opposition-based solution, as shown in Eq. (20):

$$X_i^* = 2 \times C - X_i, i = 1, 2, \dots, N-1, N \quad (20)$$

Finally, employ a greedy selection to choose the top N individuals with the best fitness values from the merged population $X \cup X^*$ as the new generation population. Here, X represents the initial population, and X^* represents the opposition-based population of the initial population. the pseudocode for MEARO is shown as Algorithm 1.

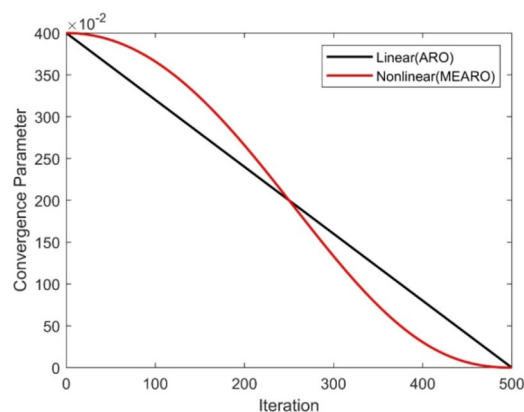


Fig. 2. Comparison of nonlinear and linear transition factors.

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1: Initialize Problem Setting ( $dim$ ,  $upper$ ,  $lower$  population ( $N$ )), iterations ( $T$ ).
2: Randomly initialize a set of Rabbits ( $X_i (i = 1, 2, \dots, N)$ ),  $X_{best}$  is the best solution found so far.
3: while  $t = 1:T$  do
4:   Randomly generate  $K$  from  $[1, N]$ 
5:   Select the top  $K$  individuals with the highest fitness from  $X$ , denoted as  $(X_1, X_2 \dots X_{K-1}, X_K)$ 
6:   Compute the centroid  $C$  using Eq. (19)
7:   For each individual  $X_i$  in  $X$  do
8:     Generate opposition-based solution  $X_i^*$  using Eq. (20)
9:   End for
8:   Merge the populations  $X_i$  and  $X_i^*$  into a combined population  $(X \cup X^*)$ 
9:   Select the top  $N$  individuals with the best fitness from  $(X \cup X^*)$  as the new population
10: End while

```

Algorithm 1. Pseudo code of SOBL.

Reinforced exploitation strategy

In optimization algorithms, using both large and small step size mutation strategies can effectively meet different search requirements. In the initial stages, a large step size accelerates global search, quickly identifying potential optimal solutions. This is particularly effective when the initial solution is far from the optimal solution. However, as the optimization process progresses, a large step size may lead to oscillations or skipping the optimal solution, especially in regions with complex solution space structures or sharp gradients. To overcome these issues, the algorithm needs to transition gradually to a smaller step size. Using a smaller step size in the later stages allows for finer parameter adjustments, accurately approaching the optimal value, and improving the stability of the optimization process.

Dynamic step size adjustment allows the algorithm to adaptively select the step size based on the current search state, enabling a balance between global search and local refinement. Therefore, to enhance the exploitation performance of the ARO algorithm, we propose a dynamic step size exploitation strategy and an improved large step size exploitation formula, setting a balancing factor ($PF = 0.5$). The enhanced exploitation strategy mathematical representation is as follows:

$$X_i(t+1) = \begin{cases} X_i(t) + (-G + 2 \times G \times r_7) \times (r_8 \times B_{i,r}(t) - X_i(t)), & \text{if } r_9 > PF \\ X_i(t) + R \times (X_{Best}(t) - r_{10} \times X_i(t)), & \text{else} \end{cases} \quad (21)$$

where, r_7 , r_8 and r_9 are random values between (0,1), r_{10} is a random integer that can be either 0 or 1. $X_{Best}(t)$ signifies the best position found by the rabbit up to the t^{th} iteration.

Based on the above discussions, the pseudocode for MEARO is shown as Algorithm 2.

```

1: Initialize Problem Setting ( $dim$ ,  $upper$ ,  $lower$  population ( $N$ )), iterations ( $T$ ).
2: Randomly initialize a set of Rabbits ( $X_i (i = 1, 2, \dots, N)$ ),  $X_{best}$  is the best solution found so far.
3: while  $t = 1:T$  do
4:   For each search agent  $X_i$  do
5:     Generate the reverse population ( $X^*$ ) using Eq. (20), and the first  $N$  individuals with the best fitness value in  $X \cup X^*$  were selected as the new generation population.
6:   Calculate the energy factor  $A$  using Eq. (18).
7:   If  $A > 1$  do
8:     Calculate the fitness  $X_{new_i}$  using Eq. (3).
9:     Update the position of the current individual using Eq. (16).
8:   else
9:     Calculate the fitness  $X_{new_i}$  using Eq. (21).
10:    Update the position of the current individual using Eq. (16).
11:   End if
12:   Update the best solution found so far  $X_{best}$ .
13: End for
14: End while
15: Return  $X_{best}$ .

```

Algorithm 2. Pseudo code of MEARO.

Algorithm complexity analysis

Analyzing the complexity of metaheuristic algorithms is vital for understanding their computational time and memory requirements. In computer science, we use “Big O notation” to describe an algorithm’s time complexity and memory usage. This study employs “Big O notation” for complexity analysis. In ARO, the complexity is expressed as: $O(ARO) = O(\text{problem definition}) + O(\text{initialization}) + O(\text{function evaluation}) + O(\text{position updates in shelter seeking}) + O(\text{position updates in random hiding})$. This simplifies to: $O(ARO) = O(1 + N + T \times N) + 0.5 \times T \times N \times dim + 0.5 \times T \times N \times dim \cong O(T \times N \times dim + T \times N + N)$. Here, N stands for the population size, dim indicates the problem’s Dimension, and T is the maximum iteration count. As this study doesn’t add new loops, the complexity stays the same as the ARO algorithm, which is $O(T \times N \times dim + T \times N + N)$.

Experiments on CEC2017 test functions and analysis

In this chapter, to confirm the efficacy of MEARO, we performed experimental verification of its performance on the CEC2017 benchmark tests. The CEC2017 benchmark functions are presented in Table 1 and fall into four categories: unimodal functions to measure precision and velocity in straightforward optimization tasks; multimodal functions to gauge local exploration capability; and hybrid and composite functions to assess handling complex continuous problems. Table 2. presents the specific parameters used for each algorithm. We employed Wilcoxon rank-sum tests to identify notable distinctions among comparative methods. Each experiment maintained a consistent sample size of 30 individuals and adhered to a predefined maximum iteration limit of 500. In order to minimize the influence of chance occurrences and potential errors, every algorithm was executed autonomously in 30 separate trials. We calculated the average and spread across these runs for comparison, highlighting the top results in bold. The experimental environment used was a Windows 11 operating system, with hardware configuration of 13th Intel(R) Core(TM) i5-13400 2.5 GHz, 16GB RAM, and MATLAB 2023b software tools.

Ablation experiment analysis

To boost ARO’s efficiency, we introduced three enhancement strategies: the Nonlinear Transition Exploration and Exploitation strategy, the Stochastic Centroid Dynamic Opposition-Based Learning strategy, and the Reinforced

Type	ID	CEC2017 function name	Range	Dimension	f_{min}
Unimodal	F1	Shifted and rotated bent cigar function	[-100,100]	30	100
	F2	Shifted and rotated sum of different power Function	[-100,100]	30	200
	F3	Shifted and rotated Zakharov function	[-100,100]	30	300
Multimodal	F4	Shifted and rotated Rosenbrock’s function	[-100,100]	30	400
	F5	Shifted and rotated Rastrigin’s function	[-100,100]	30	500
	F6	Shifted and rotated expanded Scaffer’s F6 Function	[-100,100]	30	600
	F7	Shifted and rotated Lunacek Bi_Rastrigin function	[-100,100]	30	700
	F8	Shifted and rotated non-continuous Rastrigin’s Function	[-100,100]	30	800
	F9	Shifted and rotated levy function	[-100,100]	30	900
	F10	Shifted and rotated Schwefel’s function	[-100,100]	30	1000
Hybrid	F11	Hybrid function 1 (N=3)	[-100,100]	30	1100
	F12	Hybrid function 2 (N=3)	[-100,100]	30	1200
	F13	Hybrid function 3 (N=3)	[-100,100]	30	1300
	F14	Hybrid function 4 (N=4)	[-100,100]	30	1400
	F15	Hybrid function 5 (N=4)	[-100,100]	30	1500
	F16	Hybrid function 6 (N=4)	[-100,100]	30	1600
	F17	Hybrid function 6 (N=5)	[-100,100]	30	1700
	F18	Hybrid function 6 (N=5)	[-100,100]	30	1800
	F19	Hybrid function 6 (N=5)	[-100,100]	30	1900
	F20	Hybrid function 6 (N=6)	[-100,100]	30	2000
Composition	F21	Composition function 1 (N=3)	[-100,100]	30	2100
	F22	Composition function 2 (N=3)	[-100,100]	30	2200
	F23	Composition function 3 (N=4)	[-100,100]	30	2300
	F24	Composition function 4 (N=4)	[-100,100]	30	2400
	F25	Composition function 5 (N=5)	[-100,100]	30	2500
	F26	Composition function 6 (N=5)	[-100,100]	30	2600
	F27	Composition function 7 (N=6)	[-100,100]	30	2700
	F28	Composition function 8 (N=6)	[-100,100]	30	2800
	F29	Composition function 9 (N=3)	[-100,100]	30	2900
	F30	Composition function 10 (N=3)	[-100,100]	30	3000

Table 1. CEC2017 test functions.

Algorithms	Name of the parameter	Value of the parameter
GWO	a	[0,2]
WOA	r, l, a	[0,1], [- 1,1], [0,2]
HHO	E_0, E_1	[- 1,1], [- 2,2]
PSO	c_1, c_2, w	1.49445, 1.49445, 0.9
DBO	$P_{percent}$	0.2
BKA	P, r	0.9, [0,1],
GOOSE	coe, rnd	[0,0.17], [0,1]
ARO	n_1, n_2	$N(0, 1), N(0, 1)$
MEARO	n_1, n_2, PF	$N(0, 1), N(0, 1), 0.5$

Table 2. Compare algorithm parameter settings.

Exploitation strategy. Within this segment, we explore the responsibilities and impacts of these approaches. To do so, we compared ARO variants enhanced by these strategies (NLARO, SOBLARO, and RARO) against both ARO and MEARO using the CEC2017 benchmark. Each algorithm was run independently 30 times, with performance assessed using the average and dispersion across a 30-Dimensional space. The optimal values are indicated in bold within the results presented in Table 3. Selected progression graphs for particular functions are presented in Fig. 3. The results indicate that MEARO, which combines all three approaches, delivers the top-performing solution overall. It outperformed other variant algorithms in 15 out of 30 test functions on average, underscoring the efficacy of integrating these strategies for optimal outcomes.

Swarm plots analysis

To gain a deeper understanding of MEARO’s optimization performance, we conducted iterative experiments analyzing the search history of the algorithm’s search agents. Figure 4 displays swarm plots showing the historical distribution of search agents and targets across different iteration counts. The red dots represent the targets, while the black dots represent the search agents. It is evident that at iteration 1, most search agents are far from the targets, with only a few dispersed around them. In contrast, by iteration 10, the search agents are closer to the targets, although some are still clustering in other directions, indicating convergence towards local optima. By iteration 500, the search agents are even closer to the targets and have begun to cluster together. Furthermore, there is a greater concentration of search agents around the targets, suggesting that MEARO has successfully escaped local optima and is progressing towards the global optimum.

Experimental analysis CEC2017

In this part, we evaluated the performance of MEARO against various metaheuristic methods using the CEC2017³ benchmark. This benchmark covers single-peak functions (F1-F3), straightforward multi-peak functions (F4-F10), compound functions (F11-F20), and intricate compound functions (F21-F30), all in a 30-Dimensional space. The compared algorithms include Grey Wolf Optimizer (GWO)¹⁶, Whale Optimization Algorithm (WOA)¹⁸, Harris Hawk Optimization (HHO)¹⁷, Particle Swarm Optimization (PSO)¹³, Dung Beetle Optimizer (DBO)²², Black-winged Kite Algorithm (BKA)⁶¹, GOOSE Algorithm (GOOSE)⁶², and Artificial Rabbits Optimization (ARO)⁵⁷. The parameter settings for each algorithm are detailed in Table 2. The convergence curves for different algorithms are shown Figs. 5, 6 and 7, and the test results are summarized in Tables 4, 5 and 6, including the best value (Best), worst value (Worst), average value (Ave) and standard deviation (Std). We conducted a comprehensive analysis of the data in Table 2 to determine the superior performance of MEARO. To better illustrate the advantages of each algorithm, the Friedman average ranking and final ranking values are statistically summarized in the final two rows of the table. The initial row displays the Friedman average ranking, while the subsequent row showcases the ultimate ranking for each participant. Figures 5, 6 and 7 present the convergence curves and Box plots are shown in Figs. 8, 9 and 10. For a clearer perspective on the algorithm rankings, we crafted a Sankey diagram illustrating the standings of each algorithm, depicted in Fig. 11.

In the 30 Dim evaluation, MEARO outperformed by securing the top spot in 26 functions, taking the second spot in 3 functions, and the third spot in 1 function. In 50 Dim, MEARO achieved 24 first-place rankings, 4 s-place rankings, and 2 third-place rankings. Similarly, in 100 Dim, MEARO still secured 25 first-place rankings, 2 s-place rankings, 2 third-place rankings, and 1 fourth-place ranking. This contrasts sharply with how the remaining 8 algorithms fared across the same 30 functions. Specifically, GWO, BKA, and ARO topped the list in 2, 1, and 1 function respectively. Moreover, they claimed the second spot in 7, 2, 2, and 16 functions respectively. For the third spot, GWO, HHO, PSO, DBO, BKA, GOOSE, and ARO held it in 7, 1, 7, 1, 3, 2, and 8 functions respectively. Remarkably, MEARO consistently secured a place in the top three performers across all 30 functions underscoring its stable performance. It particularly outshone in tackling unimodal and composite functions where ARO lagged, highlighting its superiority over ARO. Delving deeper into the top-three rankings, GWO exhibited a commendable consistency in securing these positions, coming close to MEARO. However, GWO’s Friedman average ranking stood at 4.30, slightly above PSO’s 4.33, suggesting its performance consistency across the 30 Dim wasn’t as steadfast as the latter two.

Across Figs. 5, 6 and 7, it becomes apparent that algorithms such as WOA, DBO, and GOOSE tend to become ensnared in local optima during later iterations, exhibiting a limited capacity to break free from them.

ID	Metric	ARO	NLARO	SOBLAEO	RARO	MEARO
F1	Ave	4.4348E+07	6.8207E+07	1.0234E+06	1.2787E+07	5.1918E+05
	Std	3.5792E+07	3.9218E+07	1.4670E+06	7.0617E+06	6.0457E+05
F2	Ave	6.1533E+22	7.4519E+22	2.7801E+17	3.0157E+22	7.8194E+16
	Std	1.9348E+23	3.5624E+23	4.9398E+17	1.4743E+23	2.1307E+17
F3	Ave	5.7338E+04	6.2081E+04	1.9436E+04	5.7174E+04	1.5332E+04
	Std	9.4335E+03	9.4637E+03	7.6987E+03	1.1241E+04	6.2008E+03
F4	Ave	5.5319E+02	5.8279E+02	5.2208E+02	5.4762E+02	5.1616E+02
	Std	6.1454E+01	5.3742E+01	2.7380E+01	3.8125E+01	2.9218E+01
F5	Ave	6.2473E+02	6.5310E+02	5.7771E+02	6.1757E+02	5.7832E+02
	Std	3.4079E+01	3.1227E+01	1.5847E+01	3.0743E+01	2.1020E+01
F6	Ave	6.1459E+02	6.2118E+02	6.0622E+02	6.1467E+02	6.0932E+02
	Std	5.7684E+00	7.9139E+00	3.2978E+00	6.2854E+00	6.1794E+00
F7	Ave	9.5655E+02	1.0041E+03	8.5033E+02	9.2397E+02	8.3306E+02
	Std	6.1237E+01	7.3996E+01	2.7025E+01	6.5595E+01	3.6489E+01
F8	Ave	9.1097E+02	9.1900E+02	8.7042E+02	8.9494E+02	8.6758E+02
	Std	2.5312E+01	2.0561E+01	1.9421E+01	2.5572E+01	1.8226E+01
F9	Ave	3.0559E+03	3.6598E+03	1.6166E+03	2.4689E+03	1.3955E+03
	Std	8.7015E+02	7.3748E+02	3.8604E+02	7.2468E+02	2.7844E+02
F10	Ave	4.6537E+03	4.6650E+03	4.3989E+03	4.4008E+03	4.5965E+03
	Std	6.5380E+02	6.7950E+02	5.8995E+02	6.1896E+02	6.6780E+02
F11	Ave	1.3155E+03	1.5109E+03	1.2341E+03	1.2874E+03	1.2456E+03
	Std	7.5082E+01	2.8155E+02	5.7915E+01	5.8824E+01	4.6604E+01
F12	Ave	3.6858E+06	4.8694E+06	3.1305E+06	3.7718E+06	2.3021E+06
	Std	2.9189E+06	5.0582E+06	2.6592E+06	3.7701E+06	1.5395E+06
F13	Ave	2.1362E+04	1.5676E+04	1.5441E+04	1.7066E+04	1.1442E+04
	Std	1.6056E+04	1.3946E+04	1.1383E+04	1.2614E+04	1.1317E+04
F14	Ave	7.7072E+04	5.4769E+04	5.7559E+04	7.1074E+04	2.6047E+04
	Std	8.1463E+04	5.3262E+04	1.0704E+05	9.0791E+04	3.8706E+04
F15	Ave	6.0335E+03	6.7201E+03	4.4917E+03	6.2612E+03	6.3919E+03
	Std	4.0587E+03	2.2241E+03	4.9613E+03	6.9264E+03	6.9028E+03
F16	Ave	2.7114E+03	2.6513E+03	2.5492E+03	2.6658E+03	2.6453E+03
	Std	3.2625E+02	2.4970E+02	2.5935E+02	2.4627E+02	2.7449E+02
F17	Ave	2.0954E+03	2.1912E+03	2.0750E+03	2.1301E+03	2.0786E+03
	Std	2.0242E+02	1.6919E+02	1.6182E+02	2.0047E+02	2.0071E+02
F18	Ave	5.7484E+05	7.2189E+05	1.2396E+05	6.8245E+05	2.1764E+05
	Std	5.8626E+05	8.4435E+05	1.1765E+05	8.4156E+05	2.8221E+05
F19	Ave	9.5694E+03	7.9546E+03	6.8767E+03	8.0255E+03	7.2714E+03
	Std	1.0390E+04	8.2767E+03	6.3623E+03	6.5266E+03	5.7708E+03
F20	Ave	2.4463E+03	2.5556E+03	2.3905E+03	2.5290E+03	2.4721E+03
	Std	2.0417E+02	1.5697E+02	1.9144E+02	1.8582E+02	1.7747E+02
F21	Ave	2.4034E+03	2.4581E+03	2.3636E+03	2.3982E+03	2.3740E+03
	Std	3.1312E+01	3.0213E+01	1.7316E+01	2.7562E+01	2.3215E+01
F22	Ave	2.3393E+03	3.0135E+03	2.3126E+03	2.4444E+03	2.5427E+03
	Std	2.4417E+01	1.4986E+03	9.7949E+00	6.3823E+02	9.1798E+02
F23	Ave	2.7955E+03	2.8236E+03	2.7511E+03	2.8068E+03	2.7660E+03
	Std	3.3517E+01	4.7165E+01	2.7667E+01	3.2567E+01	3.2177E+01
F24	Ave	2.9574E+03	3.0019E+03	2.9358E+03	2.9661E+03	2.9386E+03
	Std	4.1643E+01	5.3812E+01	4.1837E+01	4.7353E+01	3.5079E+01
F25	Ave	2.9612E+03	2.9920E+03	2.9182E+03	2.9519E+03	2.9190E+03
	Std	2.9946E+01	4.4460E+01	2.1420E+01	2.7179E+01	1.9466E+01
F26	Ave	5.2767E+03	5.1405E+03	4.9708E+03	5.2472E+03	4.9568E+03
	Std	1.2252E+03	1.5408E+03	6.5139E+02	8.2157E+02	7.4966E+02
F27	Ave	3.2828E+03	3.2881E+03	3.2067E+03	3.2818E+03	3.2049E+03
	Std	3.0533E+01	3.6844E+01	1.7408E+01	2.7987E+01	1.8445E+01
F28	Ave	3.3411E+03	3.3889E+03	3.2844E+03	3.3386E+03	3.2714E+03
	Std	3.9268E+01	6.0200E+01	2.7190E+01	2.8032E+01	2.6030E+01
Continued						

ID	Metric	ARO	NLARO	SOBLAEO	RARO	MEARO
F29	Ave	3.8955E+03	3.9638E+03	3.8007E+03	3.9282E+03	3.8586E+03
	Std	1.9802E+02	1.9829E+02	2.0129E+02	2.3840E+02	2.2174E+02
F30	Ave	9.4005E+04	1.1748E+05	4.7976E+04	7.1764E+04	2.6632E+04
	Std	8.4257E+04	1.4912E+05	4.5756E+04	6.3312E+04	2.4900E+04
Mean		3.70	4.53	1.67	3.50	1.60
Ranking		4	5	2	3	1

Table 3. Results of different strategy variants in CEC2017 experiment (dim = 30).

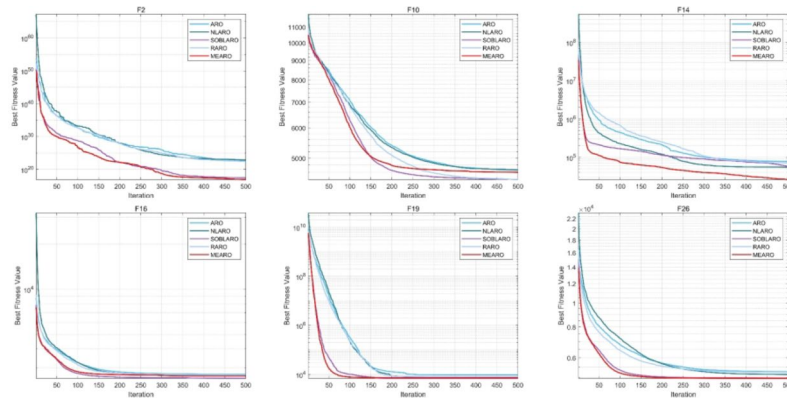


Fig. 3. Convergence curves for ARO variants and MEARO.

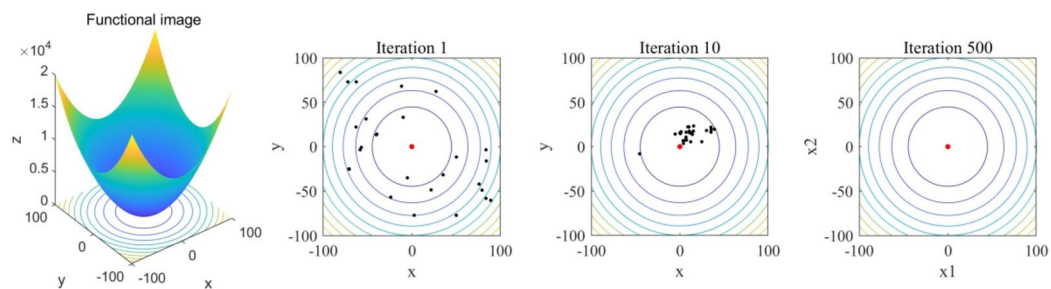


Fig. 4. Swarm plots.

Conversely, MEARO maintains excellent exploration capability even in the later iterations, successfully escaping from local optima and continuing deep exploration, ultimately attaining a high level of convergence accuracy. This indicates that our three improvement strategies are effective. These mechanisms not only facilitate the algorithm in escaping local optima but also bolster the speed and precision of convergence.

It can be clearly seen from Figs. 8, 9 and 10 that most of the MEARO box plots are the narrowest, indicating that MEARO has higher robustness compared to other algorithms. Additionally, the boxes are located near the optimal function values, suggesting that MEARO achieves greater accuracy in problem-solving compared to other algorithms. This also demonstrates the effectiveness of the three strategies embedded in ARO, which enhance the overall performance of the algorithm.

From Fig. 11, it's clear that the MEARO proposed in this study predominantly ranks first across the 30 test functions of CEC2017, with none ranking below third. By contrast, the original RAO algorithm only secured a few top rankings, underscoring the effectiveness of our enhancement strategies in bolstering the algorithm's overall performance.

Experimental analysis CEC2022

To further evaluate the scalability of SBOA, this section will compare it with other benchmark algorithms using the CEC2022 test functions⁶³ in 20 dimensions. Similar to the CEC2017 test functions, the CEC2022 test functions comprise unimodal, multimodal, hybrid, and composite functions.

Figure 12 illustrates the convergence speed of MEARO on this test suite, demonstrating that MEARO rapidly converges to the optimal solutions for most functions, outperforming ARO in both speed and accuracy.

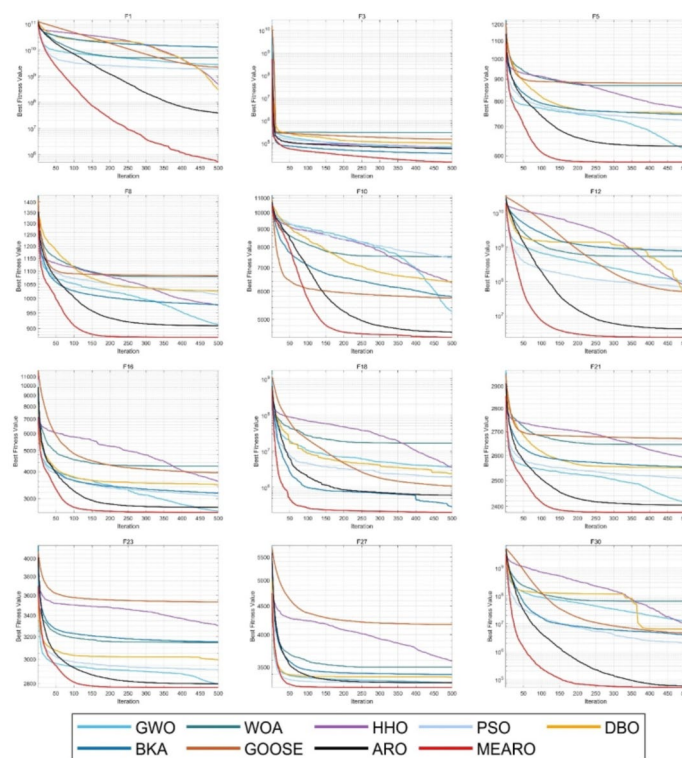


Fig. 5. CEC2017 experimental convergence curve (Dim = 30).

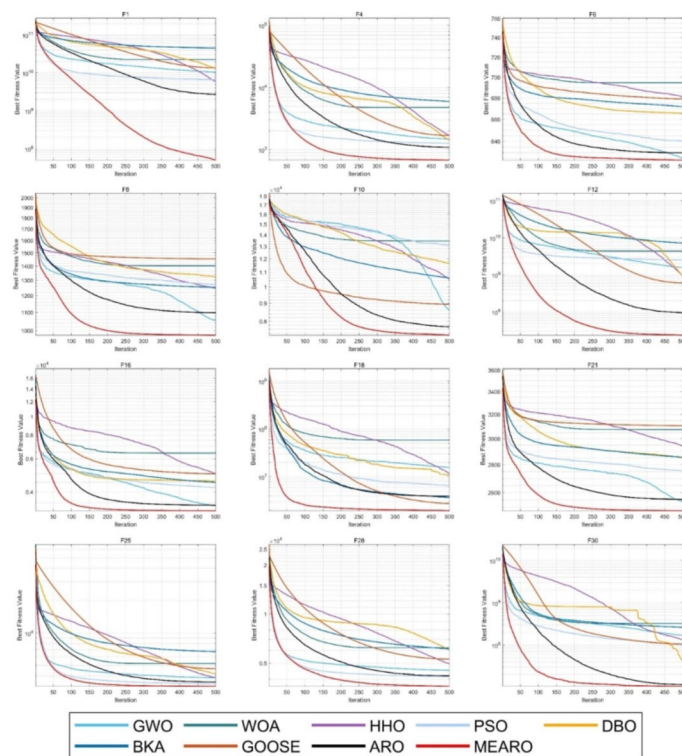


Fig. 6. CEC2017 experimental convergence curve (Dim = 50).

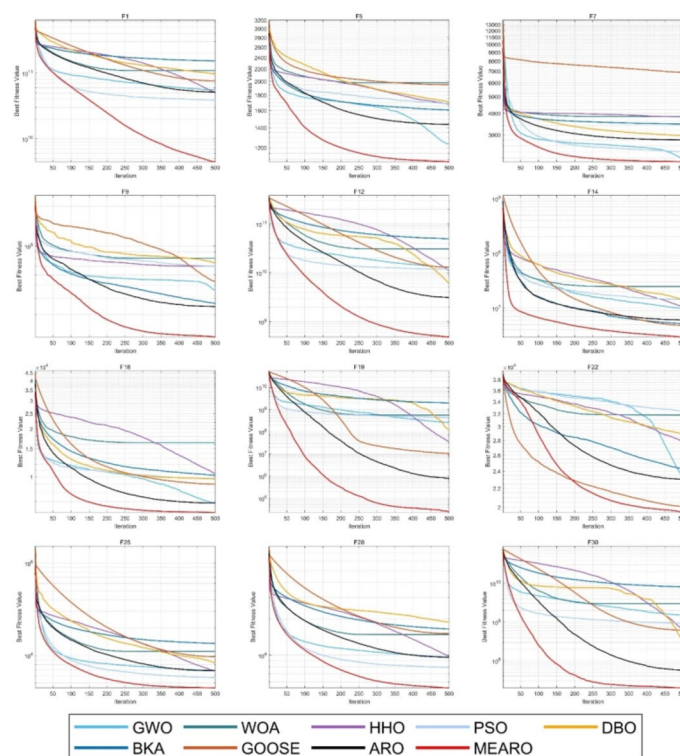


Fig. 7. CEC2017 experimental convergence curve (Dim = 100).

Table 7 presents the fitness value analysis results from 30 independent trials for each algorithm when solving the CEC2022 test functions. It is evident from the table that MEARO ranks first in 11 test functions and second in one, with no worst rankings. This performance categorizes MEARO as a high-performance optimizer, showcasing outstanding rankings across the majority of the CEC2022 test functions. Furthermore, Fig. 13 confirms that MEARO exhibits the best stability among all the compared algorithms.

Statistical analysis

To comprehensively showcase the superior performance of the proposed algorithm, this section will utilize the Wilcoxon rank-sum test to assess whether there exists a significant difference, at a significance level of $p = 5\%$, between the outcomes of each run of MEARO and those of the other algorithms. The null hypothesis H_0 posits that there is no notable distinction between the two algorithms. When $P < 5\%$, the null hypothesis is rejected, indicating a notable discrepancy between the two algorithms; when $P > 5\%$, the null hypothesis is accepted, suggesting that the variance between the two algorithms is negligible, implying comparable algorithmic performance^{2,64,65}. The symbols “+/-” are used to describe the performance comparison between MEARO and its competitors, indicating whether MEARO performs better, similarly, or worse. These data have been detailedly reported in Table 8.

By observing the statistical results, it is evident that there are significant differences between MEARO and other competing algorithms in the vast majority of cases. The specific statistical results are as follows: in the 30-Dimensional CEC2017, MEARO outperformed the comparison in 223 instances; in the 50-Dimensional case, it surpassed the comparison algorithm 221 times; and in the 100-Dimensional scenario, it excelled over the comparison algorithm 230 times. This implies that MEARO consistently achieves better results in these scenarios. Additionally, there were no instances where MEARO performed similarly to other algorithms across all three Dim. Therefore, based on the analysis of these statistical results, it can be conclusively inferred that the proposed MEARO method is significantly superior to ARO and other competing algorithms in terms of performance. This superiority is demonstrated in overall performance, making the MEARO method the most outstanding choice. Additionally, when contrasting MEARO's performance on the CEC2017 and CEC2022 test set with its counterparts and employing the Wilcoxon rank-sum test for analysis, MEARO not only demonstrates superior performance across diverse Dim but also displays extensive adaptability. This adaptability can be credited to three innovative strategies integrated into the MEARO algorithm, equipping it to outperform in a wide range of optimization challenges.

KELM improved by MEARO used for classification prediction

The KELM model

The KELM model is an evolved form of the ELM model, rooted in Single Hidden Layer Feedforward Neural Networks (SLFNs)⁶⁶. Instead of using a standard approach like its forerunner, KELM utilizes a kernel matrix

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F1	Best	2.5331E+08	2.8554E+09	8.3264E+07	4.0030E+08	9.6604E+05	3.0704E+09	1.2248E+04	8.6942E+06	4.8164E+04
	Worst	8.9035E+09	1.0679E+10	1.0961E+09	5.6414E+09	1.3137E+09	4.8825E+10	1.2298E+10	1.3219E+08	1.8779E+06
	Ave	3.4598E+09	5.5080E+09	4.0011E+08	1.4913E+09	3.3753E+08	8.6082E+09	1.7081E+09	3.8075E+07	4.5689E+05
	Std	2.2501E+09	2.1232E+09	2.5328E+08	1.3082E+09	3.0299E+08	8.1332E+09	2.6407E+09	3.0331E+07	4.3589E+05
F2	Best	7.2925E+18	6.3820E+28	2.0072E+22	3.9448E+19	4.6137E+21	5.5031E+17	7.4325E+24	6.1212E+16	6.5503E+10
	Worst	1.5440E+34	3.3518E+39	4.2179E+33	1.5690E+36	1.5885E+35	2.5483E+40	1.7856E+38	2.3148E+23	3.0280E+18
	Ave	6.0736E+32	1.1825E+38	2.6740E+32	1.0199E+35	6.8194E+33	8.4942E+38	7.1294E+36	1.4307E+22	1.8233E+17
	Std	2.8226E+33	6.1134E+38	9.1124E+32	3.8826E+35	2.9885E+34	4.6525E+39	3.2886E+37	4.6009E+22	5.6908E+17
F3	Best	3.5058E+04	1.1513E+05	3.8530E+04	2.5921E+04	7.1507E+04	2.0440E+04	5.9154E+04	4.6585E+04	6.1990E+03
	Worst	8.4155E+04	4.1246E+05	6.7146E+04	1.1485E+05	1.9514E+05	7.3849E+04	2.2257E+05	7.3907E+04	2.6903E+04
	Ave	6.5304E+04	2.6234E+05	5.9070E+04	6.4037E+04	9.4907E+04	3.3833E+04	1.3862E+05	5.9539E+04	1.4951E+04
	Std	1.0796E+04	6.0348E+04	6.3297E+03	2.1441E+04	2.6056E+04	1.2808E+04	4.0466E+04	7.1672E+03	5.6860E+03
F4	Best	5.3760E+02	7.8907E+02	5.8395E+02	5.2396E+02	5.4447E+02	5.3475E+02	5.1677E+02	5.1763E+02	4.1226E+02
	Worst	1.1249E+03	2.2496E+03	1.0953E+03	1.5607E+03	1.6944E+03	1.2785E+04	1.8796E+03	6.5974E+02	6.1287E+02
	Ave	6.4467E+02	1.3711E+03	7.4912E+02	6.7205E+02	6.9836E+02	1.9364E+03	7.3260E+02	5.5449E+02	5.1596E+02
	Std	1.2982E+02	4.0990E+02	1.2888E+02	2.2528E+02	2.2556E+02	2.8059E+03	3.2687E+02	3.2205E+01	3.4053E+01
F5	Best	5.7262E+02	7.5629E+02	6.7356E+02	6.5860E+02	6.5159E+02	6.7065E+02	7.6069E+02	5.7008E+02	5.3588E+02
	Worst	7.6074E+02	1.0222E+03	8.3491E+02	7.8960E+02	8.6390E+02	9.2549E+02	1.0325E+03	6.7489E+02	6.2800E+02
	Ave	6.2933E+02	8.6006E+02	7.8053E+02	7.1009E+02	7.5030E+02	7.5893E+02	8.7002E+02	6.1996E+02	5.7240E+02
	Std	4.9774E+01	6.4319E+01	3.0750E+01	3.0830E+01	4.7658E+01	5.1336E+01	6.8591E+01	2.5854E+01	1.7819E+01
F6	Best	6.0359E+02	6.6031E+02	6.5070E+02	6.1178E+02	6.2559E+02	6.4351E+02	6.6166E+02	6.0489E+02	6.0216E+02
	Worst	6.2439E+02	7.0436E+02	6.7920E+02	6.3628E+02	6.7769E+02	6.9652E+02	6.9246E+02	6.2571E+02	6.1662E+02
	Ave	6.1383E+02	6.8204E+02	6.7000E+02	6.2214E+02	6.4982E+02	6.6394E+02	6.7704E+02	6.1472E+02	6.0862E+02
	Std	4.9101E+00	1.1227E+01	5.8045E+00	5.9233E+00	1.2938E+01	1.2779E+01	8.6858E+00	5.7050E+00	4.0151E+00
F7	Best	8.3630E+02	1.0543E+03	1.1836E+03	9.2575E+02	8.4814E+02	1.1040E+03	1.3237E+03	8.4612E+02	7.8526E+02
	Worst	1.0097E+03	1.4814E+03	1.4136E+03	1.0644E+03	1.1519E+03	1.4137E+03	3.5355E+03	1.0272E+03	9.3260E+02
	Ave	9.0144E+02	1.3282E+03	1.3104E+03	9.9440E+02	9.9116E+02	1.2408E+03	2.1506E+03	9.1239E+02	8.3884E+02
	Std	4.2839E+01	9.2662E+01	6.4667E+01	3.4835E+01	8.0007E+01	7.6331E+01	8.6902E+02	3.9407E+01	3.3062E+01
F8	Best	8.6382E+02	9.4653E+02	9.4998E+02	9.6444E+02	9.5005E+02	9.1970E+02	9.8706E+02	8.7649E+02	8.4496E+02
	Worst	1.0325E+03	1.1795E+03	1.0471E+03	1.0532E+03	1.1280E+03	1.1572E+03	1.2880E+03	9.4033E+02	9.1445E+02
	Ave	9.0676E+02	1.0655E+03	9.8820E+02	1.0176E+03	1.0268E+03	9.8973E+02	1.0968E+03	9.0410E+02	8.6725E+02
	Std	3.3088E+01	4.9348E+01	2.1407E+01	2.2709E+01	4.6044E+01	5.3979E+01	7.7143E+01	1.7488E+01	1.4110E+01
F9	Best	1.1515E+03	5.8823E+03	7.1029E+03	1.3134E+03	3.6842E+03	4.7129E+03	5.4909E+03	1.5131E+03	1.0846E+03
	Worst	3.5656E+03	1.7080E+04	1.1699E+04	8.7915E+03	1.0131E+04	8.6013E+03	1.2836E+04	4.6593E+03	3.0762E+03
	Ave	2.3564E+03	1.0412E+04	8.9723E+03	2.0806E+03	6.5734E+03	5.4882E+03	8.1002E+03	2.6805E+03	1.5836E+03
	Std	6.8714E+02	2.8493E+03	1.0660E+03	1.4942E+03	1.7371E+03	7.3484E+02	2.5087E+03	8.4755E+02	4.4045E+02
F10	Best	3.1389E+03	5.7586E+03	4.6672E+03	6.1009E+03	4.3112E+03	3.9391E+03	4.4514E+03	3.4638E+03	2.4998E+03
	Worst	9.0205E+03	9.0796E+03	7.8013E+03	8.5790E+03	9.0459E+03	9.1248E+03	7.0796E+03	5.5030E+03	5.3579E+03
	Ave	5.0506E+03	7.3693E+03	5.9236E+03	7.4777E+03	6.5039E+03	5.5945E+03	5.7408E+03	4.4498E+03	4.3608E+03
	Std	1.1986E+03	7.3956E+02	7.7294E+02	5.6378E+02	1.1069E+03	1.1575E+03	7.8575E+02	5.7646E+02	6.3662E+02
F11	Best	1.3730E+03	2.8195E+03	1.2432E+03	1.3192E+03	1.3370E+03	1.2646E+03	1.2234E+03	1.2069E+03	1.1624E+03
	Worst	4.6793E+03	1.6002E+04	1.8674E+03	1.6985E+03	4.7292E+03	1.0171E+04	8.7094E+03	2.5124E+03	1.3131E+03
	Ave	2.4166E+03	9.7949E+03	1.5433E+03	1.4878E+03	1.9836E+03	1.9817E+03	1.9336E+03	1.3829E+03	1.2438E+03
	Std	1.0725E+03	3.2463E+03	1.4969E+02	8.9773E+01	7.3730E+02	1.7896E+03	1.3626E+03	2.3976E+02	4.1859E+01
F12	Best	7.9511E+06	6.3563E+07	8.1072E+06	2.1166E+07	3.6854E+06	3.8557E+06	5.7020E+05	7.1462E+05	3.8026E+05
	Worst	2.8740E+08	1.1201E+09	3.2434E+08	3.1503E+08	5.8172E+08	7.8343E+09	1.2685E+08	9.1159E+06	5.0998E+06
	Ave	8.3972E+07	3.8059E+08	9.0004E+07	7.6654E+07	7.4069E+07	1.0331E+09	2.7528E+07	3.3443E+06	2.2518E+06
	Std	7.1182E+07	2.5260E+08	6.9994E+07	6.5052E+07	1.1171E+08	2.4203E+09	3.0769E+07	2.2479E+06	1.3771E+06
F13	Best	4.8047E+04	8.5825E+05	1.7603E+05	2.1582E+05	2.5288E+04	5.5417E+04	2.3987E+04	2.7409E+03	1.9643E+03
	Worst	1.0788E+08	1.0972E+08	2.5188E+06	7.2455E+07	7.1333E+07	4.1966E+09	6.5558E+05	6.4885E+04	3.8663E+04
	Ave	7.0359E+06	1.3625E+07	1.0519E+06	7.1115E+06	8.8074E+06	1.9911E+08	1.5451E+05	1.8510E+04	1.2892E+04
	Std	2.0959E+07	2.0598E+07	5.4111E+05	1.2930E+07	1.5345E+07	7.8263E+08	1.3599E+05	1.6937E+04	8.4520E+03
F14	Best	2.3033E+04	5.9082E+04	1.1702E+05	6.6808E+03	8.9558E+03	1.8232E+03	3.4722E+03	3.2144E+03	1.5552E+03
	Worst	2.2998E+06	1.1504E+07	5.1908E+06	3.4138E+05	6.9242E+05	3.6177E+05	1.2334E+06	4.6698E+05	1.7872E+05
	Ave	7.3650E+05	2.6635E+06	1.5103E+06	1.0544E+05	2.1033E+05	2.2066E+04	1.8409E+05	7.7775E+04	2.3627E+04
	Std	7.3220E+05	3.0603E+06	1.2191E+06	7.1106E+04	1.9370E+05	6.6569E+04	2.9306E+05	9.3534E+04	3.5420E+04
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F15	Best	2.6416E+04	2.0148E+05	3.6422E+04	1.6193E+04	9.5176E+03	5.5741E+03	8.4076E+03	1.8089E+03	1.5479E+03
	Worst	5.8521E+06	4.6992E+07	3.6691E+05	6.9128E+05	1.2608E+06	5.9057E+05	1.5314E+05	1.5118E+04	2.2912E+04
	Ave	1.3768E+06	9.4114E+06	1.2277E+05	1.6656E+05	1.4132E+05	4.6452E+04	5.1890E+04	5.1368E+03	5.8034E+03
	Std	1.8018E+06	1.2164E+07	7.6889E+04	1.5656E+05	2.3503E+05	1.0606E+05	4.3014E+04	3.5435E+03	5.5549E+03
F16	Best	2.1703E+03	2.8277E+03	2.6323E+03	2.1215E+03	2.2486E+03	2.3864E+03	2.8909E+03	2.1090E+03	2.0357E+03
	Worst	3.6824E+03	5.9166E+03	4.6490E+03	3.7726E+03	4.4995E+03	4.0636E+03	6.0285E+03	3.2997E+03	3.1170E+03
	Ave	2.7404E+03	4.1959E+03	3.5245E+03	2.9616E+03	3.3737E+03	2.9983E+03	4.0025E+03	2.7077E+03	2.5837E+03
	Std	3.9715E+02	7.4830E+02	4.2557E+02	3.9922E+02	5.5306E+02	3.6993E+02	6.4816E+02	2.6651E+02	2.7219E+02
F17	Best	1.8027E+03	2.2668E+03	2.1828E+03	1.8984E+03	2.2879E+03	1.8406E+03	2.3302E+03	1.7926E+03	1.7843E+03
	Worst	2.4262E+03	3.3372E+03	3.4045E+03	2.6287E+03	3.1285E+03	2.9794E+03	3.8003E+03	2.6275E+03	2.6246E+03
	Ave	2.0798E+03	2.7519E+03	2.7323E+03	2.1938E+03	2.6973E+03	2.3064E+03	2.8726E+03	2.1459E+03	2.0777E+03
	Std	1.8127E+02	2.9008E+02	2.7897E+02	1.9170E+02	2.5188E+02	2.4637E+02	3.5583E+02	2.1857E+02	1.8484E+02
F18	Best	8.9426E+04	2.4669E+05	4.9570E+05	1.3296E+05	1.0548E+05	5.4894E+04	1.1701E+05	5.4687E+04	2.0942E+04
	Worst	2.7645E+07	9.2675E+07	2.3232E+07	4.9650E+06	1.7166E+07	1.1320E+07	4.4288E+06	5.6589E+06	3.6701E+06
	Ave	3.0795E+06	1.6867E+07	5.2404E+06	1.6952E+06	3.1635E+06	6.1138E+05	1.4935E+06	6.2422E+05	3.9610E+05
	Std	5.3404E+06	1.9993E+07	5.1502E+06	1.2313E+06	4.2843E+06	2.0967E+06	1.2615E+06	1.1722E+06	6.8449E+05
F19	Best	1.4065E+04	1.3447E+06	1.6815E+05	3.0899E+04	2.6156E+03	7.9395E+03	1.4665E+05	1.9384E+03	1.9389E+03
	Worst	3.1391E+06	6.0885E+07	2.7379E+06	2.1086E+06	1.4890E+07	3.5837E+07	2.9301E+06	4.6741E+04	1.5315E+04
	Ave	9.4426E+05	2.1628E+07	1.2452E+06	3.7219E+05	2.2655E+06	1.5363E+06	1.2592E+06	9.4907E+03	6.6433E+03
	Std	8.6916E+05	1.6068E+07	6.9590E+05	4.6881E+05	3.9479E+06	6.4890E+06	7.5597E+05	1.0291E+04	3.9939E+03
F20	Best	2.1618E+03	2.4239E+03	2.3611E+03	2.1775E+03	2.2923E+03	2.3125E+03	2.6001E+03	2.0654E+03	2.0626E+03
	Worst	2.8293E+03	3.5815E+03	3.2295E+03	3.1566E+03	3.1659E+03	2.9910E+03	3.7507E+03	2.9616E+03	2.8292E+03
	Ave	2.4935E+03	2.9402E+03	2.8013E+03	2.5107E+03	2.7658E+03	2.5670E+03	3.1224E+03	2.5217E+03	2.4266E+03
	Std	1.7662E+02	2.2523E+02	2.2661E+02	2.2795E+02	1.9473E+02	1.6598E+02	2.4970E+02	2.0134E+02	2.0221E+02
F21	Best	2.3557E+03	2.5388E+03	2.5122E+03	2.4525E+03	2.4882E+03	2.4424E+03	2.5405E+03	2.3597E+03	2.3339E+03
	Worst	2.5429E+03	2.8124E+03	2.6825E+03	2.5648E+03	2.6426E+03	2.7216E+03	2.8480E+03	2.4540E+03	2.4182E+03
	Ave	2.4009E+03	2.6566E+03	2.5901E+03	2.5052E+03	2.5664E+03	2.5481E+03	2.6777E+03	2.4012E+03	2.3780E+03
	Std	3.3177E+01	6.0921E+01	5.0296E+01	2.9303E+01	4.1473E+01	6.7320E+01	8.3143E+01	2.5332E+01	2.0559E+01
F22	Best	2.4798E+03	4.3859E+03	5.5578E+03	2.4671E+03	2.3575E+03	3.5772E+03	3.9207E+03	2.3203E+03	2.3025E+03
	Worst	1.0437E+04	9.6803E+03	8.3432E+03	9.8131E+03	8.8849E+03	9.5698E+03	8.7952E+03	6.2682E+03	6.8341E+03
	Ave	5.4974E+03	8.6632E+03	7.2610E+03	5.7352E+03	5.0990E+03	6.5447E+03	7.3635E+03	2.4708E+03	2.7032E+03
	Std	2.2706E+03	9.4230E+02	8.1184E+02	3.0676E+03	2.6251E+03	1.3404E+03	1.1129E+03	7.1757E+02	1.2112E+03
F23	Best	2.7370E+03	2.9492E+03	3.0130E+03	2.8450E+03	2.8809E+03	2.9669E+03	3.2157E+03	2.7367E+03	2.7038E+03
	Worst	2.9465E+03	3.3597E+03	3.5155E+03	3.0651E+03	3.1857E+03	3.2923E+03	3.7757E+03	2.9015E+03	2.8107E+03
	Ave	2.7819E+03	3.1826E+03	3.2551E+03	2.9354E+03	3.0093E+03	3.1030E+03	3.5172E+03	2.7880E+03	2.7609E+03
	Std	4.6687E+01	1.1052E+02	1.3016E+02	5.0609E+01	7.5722E+01	9.8853E+01	1.6608E+02	4.0216E+01	2.5969E+01
F24	Best	2.8887E+03	3.1090E+03	3.1921E+03	3.0039E+03	3.0136E+03	3.0415E+03	3.3222E+03	2.8818E+03	2.8735E+03
	Worst	3.1365E+03	3.5139E+03	3.7826E+03	3.3039E+03	3.4219E+03	3.5183E+03	3.9659E+03	3.0549E+03	3.0534E+03
	Ave	2.9715E+03	3.2633E+03	3.4950E+03	3.1439E+03	3.1725E+03	3.2566E+03	3.6739E+03	2.9569E+03	2.9469E+03
	Std	7.3739E+01	1.0719E+02	1.4202E+02	8.1469E+01	9.2729E+01	1.1670E+02	1.3524E+02	3.6591E+01	4.4197E+01
F25	Best	2.9080E+03	3.0918E+03	2.9489E+03	2.9160E+03	2.8959E+03	2.9274E+03	2.9025E+03	2.8966E+03	2.8881E+03
	Worst	3.2165E+03	3.4029E+03	3.1047E+03	3.0374E+03	3.1704E+03	4.5674E+03	3.4383E+03	3.0245E+03	2.9445E+03
	Ave	3.0132E+03	3.2283E+03	3.0096E+03	2.9727E+03	2.9909E+03	3.1855E+03	3.0292E+03	2.9626E+03	2.9109E+03
	Std	6.0005E+01	8.2655E+01	3.3661E+01	2.8969E+01	6.9233E+01	3.2731E+02	1.2399E+02	2.5301E+01	1.9374E+01
F26	Best	4.1849E+03	7.1524E+03	7.1082E+03	3.2448E+03	3.8299E+03	3.7410E+03	6.5996E+03	2.9455E+03	2.8101E+03
	Worst	6.0710E+03	1.0311E+04	1.0085E+04	7.0781E+03	8.5749E+03	1.0855E+04	1.6942E+04	7.3313E+03	5.9292E+03
	Ave	4.9558E+03	8.7453E+03	8.5237E+03	4.9780E+03	7.0498E+03	7.5489E+03	1.0711E+04	4.8595E+03	5.0179E+03
	Std	4.3872E+02	8.3708E+02	8.0172E+02	1.1401E+03	8.8156E+02	1.8355E+03	2.7467E+03	1.4200E+03	7.2894E+02
F27	Best	3.2158E+03	3.2657E+03	3.3406E+03	3.2174E+03	3.2477E+03	3.2484E+03	3.4431E+03	3.2330E+03	3.2000E+03
	Worst	3.3542E+03	4.1175E+03	4.5108E+03	3.4089E+03	3.4624E+03	4.3052E+03	5.1660E+03	3.3336E+03	3.3106E+03
	Ave	3.2599E+03	3.5325E+03	3.6626E+03	3.2741E+03	3.3304E+03	3.4605E+03	4.1140E+03	3.2759E+03	3.2170E+03
	Std	2.7672E+01	2.0244E+02	2.9036E+02	4.1386E+01	4.9048E+01	2.3271E+02	4.3770E+02	2.7065E+01	2.9820E+01
F28	Best	3.3289E+03	3.5645E+03	3.3147E+03	3.3019E+03	3.3217E+03	3.3613E+03	3.3136E+03	3.2808E+03	3.2095E+03
	Worst	3.6907E+03	4.1469E+03	3.7006E+03	3.5304E+03	6.5864E+03	5.3900E+03	5.2614E+03	3.4206E+03	3.3233E+03
	Ave	3.4736E+03	3.8382E+03	3.5146E+03	3.3656E+03	3.5431E+03	3.7065E+03	3.6648E+03	3.3365E+03	3.2717E+03
	Std	9.8550E+01	1.5916E+02	1.1062E+02	5.1707E+01	5.7814E+02	4.2920E+02	5.0294E+02	3.2686E+01	2.9540E+01
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F29	Best	3.6480E+03	4.1945E+03	4.0094E+03	3.6715E+03	3.9679E+03	3.6833E+03	4.1491E+03	3.5054E+03	3.4222E+03
	Worst	4.3913E+03	7.2437E+03	6.7112E+03	4.8392E+03	5.0746E+03	6.5594E+03	6.5797E+03	4.4043E+03	3.9809E+03
	Ave	3.9147E+03	5.4204E+03	5.0537E+03	4.0877E+03	4.4766E+03	4.7126E+03	5.3866E+03	3.8461E+03	3.7304E+03
	Std	1.4941E+02	6.5686E+02	5.3909E+02	2.8606E+02	3.0385E+02	5.3104E+02	5.6673E+02	2.6882E+02	1.3853E+02
F30	Best	2.8249E+06	7.9484E+06	9.5986E+05	1.5576E+05	1.1811E+04	2.1211E+05	1.7429E+05	8.8059E+03	9.2575E+03
	Worst	3.2472E+07	2.7509E+08	1.2181E+08	5.7030E+06	3.7167E+07	1.0110E+08	2.4041E+07	2.7737E+05	1.0080E+06
	Ave	1.2678E+07	7.4429E+07	1.7162E+07	1.9804E+06	4.1904E+06	1.1491E+07	5.1675E+06	5.7713E+04	9.1334E+04
	Std	8.0107E+06	6.6113E+07	2.4374E+07	1.6255E+06	8.1497E+06	2.4216E+07	4.7660E+06	5.9764E+04	1.8351E+05
Mean		4.30	8.47	6.77	4.33	5.40	5.10	6.87	2.60	1.17
Ranking		3	9	7	4	6	5	8	2	1

Table 4. Experimental evaluation outcomes utilizing CEC2017 test functions (dim = 30).

for the hidden layer's feature transformation. Emerging as a cutting-edge method, KELM has shown superior generalization capabilities compared to ELM in various real-world scenarios by leveraging kernel functions^{67,68}. To facilitate understanding, this section briefly describes the characteristics of KELM.

ELM's core principle revolves around finding distinct optimal solutions by randomly setting input weights and hidden layer biases in a neural network, as opposed to fine-tuning them. With N training samples (x_i, t_i) , K hidden neurons, and an activation function $g(x)$, where each input x_i is an $n \times 1$ Dimensional vector $x_i \in R^n$ and t_i is an $m \times 1$ Dimensional target vector $t_i \in R^m$, the output of the ELM is determined by a linear system. The mathematical representation of SLFNs is articulated as^{69,70}:

$$O_j = \sum_{i=1}^K \beta_i g(w_i \times x_i + b_i) \quad i = 1, 2, \dots, N; j = 1, 2, \dots, N \quad (22)$$

Here O_j represents the output vector for the j^{th} input, b_i stands for the vector of output weights that links the i^{th} hidden layer to the output layer, w_i the vector of weights that connects the i^{th} hidden neuron to the input layer, the notation $w_i \times x_i$ signifies the dot product of w_i and x_i , and $g(w_i \times x_i + b_i)$ represents the activation function for the i^{th} hidden neuron within the hidden layer.

The goal of ELM's learning process is to reduce the training error. If SLFNs can perfectly fit the N training samples with zero error, the following equations are satisfied: $\sum_{i=1}^N \|t_i \times o_i\| = 0$, where $\beta_i w_i$ and b_i exist such that:

$$t_j = o_j = \sum_{i=1}^K \beta_i g(w_i \times x_i + b_i) \quad j = 1, 2, \dots, N \quad (23)$$

$$T = h(x) \beta = H \times \beta \quad (24)$$

Where $T = [t_1, t_2, \dots, t_N]^T$ and $b = [b_1, b_2, \dots, b_K]^T$, $h(x)$ represents the mapping function that converts data from its original input space into a feature space of Dimensionality K :

$$H = h(x) = \begin{bmatrix} h_1(x_1) \\ \vdots \\ h_K(x_1) \end{bmatrix}^T = \begin{bmatrix} g(w_1 \times x_1 + b_1) & \cdots & g(w_K \times x_1 + b_K) \\ \vdots & \ddots & \vdots \\ g(w_1 \times x_N + b_1) & \cdots & g(w_K \times x_N + b_K) \end{bmatrix}_{N \times K} \quad (25)$$

Let's say H is the matrix that contains the outputs of the hidden layer in the neural network. Each column of H corresponds to the output of a specific hidden neuron in relation to the N -Dimensional input. In training SLFNs, there's no requirement to tweak the weights or biases of the hidden layer; they can be set arbitrarily. The network's output b can be calculated using analytical methods. With this in mind, only the weights for the output need adjustments to optimize the network. Here are the guidelines for setting these output weights:

$$\beta' = H^+ \times T \quad (26)$$

Where H^+ is the Moore–Penrose generalized inverse of the hidden layer output matrix H , computed using orthogonal projection, i.e., $H^+ = H^T (HH^T)^{-1}$. Using the Moore–Penrose inverse technique ensures the smallest possible norm among all potential least squares solutions. This method can greatly enhance the learning rate and lead to robust generalization capabilities.

Kernel-based ELM was proposed by Huang et al.³³ with the aim of enhancing the generalization capability of ELM, surpassing the least squares-based ELM. It is recommended to add a regularization term, typically a constant C , to the diagonal of $H^T H$ when computing the output weights b , as follows:

$$\beta = H^T \left(\frac{I}{C} + HH^T \right)^{-1} T \quad (27)$$

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F1	Best	3.9821E+09	1.3630E+10	2.2609E+09	2.9446E+09	1.4552E+09	2.2244E+10	4.6462E+08	7.3853E+08	8.7902E+06
	Worst	1.9940E+10	3.1843E+10	1.1211E+10	1.2275E+10	8.3356E+09	1.1256E+11	4.3223E+10	5.5590E+09	2.3708E+08
	Ave	1.1436E+10	2.1197E+10	5.5682E+09	6.0768E+09	3.6373E+09	4.0376E+10	1.4465E+10	2.3967E+09	5.8820E+07
	Std	4.2730E+09	4.8670E+09	2.1663E+09	2.2912E+09	1.6560E+09	1.6867E+10	1.3538E+10	1.2352E+09	5.6069E+07
F2	Best	2.3197E+44	1.3642E+59	9.3317E+49	1.3665E+45	1.6342E+50	2.5214E+46	5.3950E+42	4.9248E+42	2.3742E+32
	Worst	1.0955E+64	3.8035E+82	1.7651E+65	1.0391E+59	6.8825E+69	4.0610E+74	8.1660E+66	1.1530E+50	1.9479E+46
	Ave	3.6517E+62	1.7984E+81	1.6309E+64	3.4637E+57	2.3017E+68	1.3537E+73	3.2485E+65	6.5915E+48	1.2289E+45
	Std	2.0001E+63	7.4317E+81	4.0737E+64	1.8971E+58	1.2564E+69	7.4143E+73	1.5070E+66	2.2219E+49	4.3139E+45
F3	Best	1.0390E+05	1.5203E+05	1.3900E+05	1.3551E+05	1.8072E+05	5.4487E+04	2.1584E+05	1.1032E+05	3.1259E+04
	Worst	2.4358E+05	6.3510E+05	2.1255E+05	3.8958E+05	6.0009E+05	2.4137E+05	4.2922E+05	2.0539E+05	1.1172E+05
	Ave	1.7045E+05	3.3014E+05	1.7418E+05	2.2560E+05	2.6839E+05	9.7732E+04	3.1425E+05	1.6333E+05	7.0392E+04
	Std	3.4536E+04	1.1556E+05	1.8099E+04	5.8212E+04	8.6963E+04	4.0063E+04	4.9269E+04	2.1712E+04	1.9307E+04
F4	Best	7.8039E+02	2.3945E+03	1.2094E+03	7.6132E+02	7.2445E+02	1.3690E+03	7.4329E+02	7.2563E+02	5.7989E+02
	Worst	4.9825E+03	7.2355E+03	4.3524E+03	2.7991E+03	4.8335E+03	2.3443E+04	9.5212E+03	1.9992E+03	8.6680E+02
	Ave	1.7304E+03	4.9070E+03	1.9576E+03	1.1807E+03	1.3575E+03	5.8641E+03	2.9643E+03	1.0940E+03	6.8422E+02
	Std	9.0369E+02	1.1283E+03	7.1449E+02	4.7250E+02	7.2462E+02	4.1077E+03	2.2190E+03	2.7520E+02	7.0139E+01
F5	Best	6.5285E+02	9.3425E+02	8.6871E+02	8.7200E+02	8.2734E+02	8.2373E+02	8.7716E+02	7.2695E+02	6.2236E+02
	Worst	1.0548E+03	1.2730E+03	1.0139E+03	1.0428E+03	1.1508E+03	1.1528E+03	1.4164E+03	8.9812E+02	7.4930E+02
	Ave	7.6711E+02	1.0876E+03	9.3542E+02	9.6026E+02	9.9989E+02	9.0891E+02	1.1041E+03	7.9521E+02	6.7961E+02
	Std	6.6858E+01	6.3629E+01	3.9177E+01	4.2702E+01	8.4912E+01	7.2837E+01	1.8760E+02	4.1799E+01	3.1879E+01
F6	Best	6.1258E+02	6.7189E+02	6.6810E+02	6.2277E+02	6.4858E+02	6.6139E+02	6.6411E+02	6.2313E+02	6.1018E+02
	Worst	6.4201E+02	7.2932E+02	6.8872E+02	6.9698E+02	6.9701E+02	7.0879E+02	6.9740E+02	6.5525E+02	6.4519E+02
	Ave	6.2661E+02	7.0143E+02	6.8034E+02	6.4087E+02	6.6780E+02	6.7427E+02	6.7913E+02	6.3142E+02	6.2383E+02
	Std	6.0419E+00	1.3571E+01	4.5780E+00	1.4673E+01	1.0228E+01	1.1223E+01	8.8832E+00	7.1471E+00	7.6601E+00
F7	Best	1.0260E+03	1.6285E+03	1.7149E+03	1.2026E+03	1.0920E+03	1.5912E+03	1.7556E+03	1.1401E+03	9.6817E+02
	Worst	1.3200E+03	2.1737E+03	2.0268E+03	1.4717E+03	1.9082E+03	2.1194E+03	5.7698E+03	1.7082E+03	1.2625E+03
	Ave	1.1575E+03	1.8886E+03	1.8665E+03	1.3205E+03	1.3956E+03	1.7654E+03	3.7934E+03	1.3758E+03	1.0742E+03
	Std	7.9069E+01	1.2077E+02	8.0731E+01	5.9138E+01	1.6774E+02	1.4887E+02	1.7469E+03	1.2459E+02	7.0675E+01
F8	Best	9.7927E+02	1.2743E+03	1.1861E+03	1.1765E+03	1.0805E+03	1.1488E+03	1.2289E+03	9.9906E+02	9.3874E+02
	Worst	1.1535E+03	1.6257E+03	1.2921E+03	1.3814E+03	1.4918E+03	1.5299E+03	1.7626E+03	1.1499E+03	1.0284E+03
	Ave	1.0577E+03	1.4018E+03	1.2354E+03	1.2608E+03	1.2781E+03	1.2387E+03	1.4340E+03	1.0978E+03	9.8173E+02
	Std	4.5399E+01	8.8311E+01	2.8874E+01	4.8459E+01	1.2521E+02	9.5529E+01	1.6931E+02	3.5096E+01	2.2676E+01
F9	Best	3.8906E+03	1.8058E+04	2.7119E+04	3.5843E+03	1.4608E+04	1.1293E+04	1.3847E+04	5.4336E+03	3.1427E+03
	Worst	1.8344E+04	7.7543E+04	3.6454E+04	3.7026E+04	4.1510E+04	3.5345E+04	3.7466E+04	1.7184E+04	8.1644E+03
	Ave	1.0287E+04	4.3135E+04	3.1851E+04	1.2575E+04	2.7034E+04	1.7507E+04	2.2380E+04	1.0443E+04	5.2816E+03
	Std	3.6488E+03	1.1842E+04	2.4994E+03	9.0435E+03	7.3347E+03	5.3017E+03	7.2750E+03	2.9628E+03	1.1472E+03
F10	Best	6.2810E+03	1.0707E+04	7.7346E+03	1.1042E+04	7.7020E+03	7.1670E+03	7.0240E+03	5.9806E+03	5.3953E+03
	Worst	1.5196E+04	1.5432E+04	1.2598E+04	1.4509E+04	1.5519E+04	1.5730E+04	1.0107E+04	9.5572E+03	8.9786E+03
	Ave	8.2037E+03	1.3264E+04	1.0213E+04	1.3143E+04	1.1732E+04	9.6793E+03	8.5760E+03	7.9661E+03	7.5728E+03
	Std	1.9727E+03	1.2168E+03	1.0459E+03	8.5613E+02	2.3867E+03	2.0055E+03	7.4242E+02	7.9811E+02	8.7340E+02
F11	Best	2.7101E+03	6.0029E+03	2.1806E+03	1.7509E+03	2.8048E+03	2.2524E+03	1.4505E+03	1.7075E+03	1.2346E+03
	Worst	1.4751E+04	1.4064E+04	5.0470E+03	7.5538E+03	1.9163E+04	2.5984E+04	1.1911E+04	4.6167E+03	1.6247E+03
	Ave	7.3331E+03	8.9329E+03	2.9649E+03	2.7425E+03	5.4142E+03	6.8183E+03	4.3234E+03	2.4137E+03	1.3935E+03
	Std	2.9957E+03	2.2564E+03	6.7621E+02	9.8000E+02	3.2217E+03	6.2943E+03	2.7745E+03	6.6223E+02	7.7914E+01
F12	Best	7.7608E+07	1.6745E+09	2.1275E+08	5.6547E+08	2.0726E+08	9.4179E+07	3.5573E+07	1.7443E+07	2.2788E+06
	Worst	4.7425E+09	9.1877E+09	5.8257E+09	8.2200E+09	3.3597E+09	5.8053E+10	2.1833E+09	1.6832E+08	4.8411E+07
	Ave	1.3223E+09	4.3513E+09	1.2137E+09	2.1783E+09	1.0076E+09	1.0421E+10	4.4133E+08	8.0117E+07	1.6527E+07
	Std	1.2479E+09	1.5616E+09	1.1456E+09	1.9470E+09	6.7153E+08	1.5049E+10	5.4371E+08	4.0674E+07	1.0972E+07
F13	Best	9.8766E+06	1.4580E+08	5.8978E+06	2.4449E+07	5.8489E+05	1.2401E+06	2.6960E+04	3.8415E+04	2.6918E+03
	Worst	1.0959E+09	1.2590E+09	1.1926E+08	3.0975E+09	2.3201E+08	3.1265E+10	4.1433E+05	8.4853E+05	1.1082E+05
	Ave	2.4290E+08	5.2472E+08	3.0445E+07	6.7680E+08	8.4887E+07	1.4587E+09	1.5552E+05	2.3783E+05	1.7469E+04
	Std	2.1875E+08	2.7684E+08	2.5214E+07	9.1086E+08	6.4008E+07	5.9242E+09	1.1262E+05	1.9648E+05	2.1045E+04
F14	Best	1.6340E+05	6.1136E+05	1.3036E+06	2.4607E+05	1.4446E+05	3.6990E+04	4.9988E+04	7.0070E+04	2.8304E+04
	Worst	8.3039E+06	2.7159E+07	2.2073E+07	8.6658E+06	1.8032E+07	1.1775E+07	2.5783E+06	3.3679E+06	1.1330E+06
	Ave	1.6065E+06	9.0338E+06	6.1804E+06	1.3659E+06	3.6701E+06	7.6244E+05	5.1342E+05	9.9282E+05	2.3811E+05
	Std	2.0566E+06	6.3785E+06	4.7333E+06	1.6257E+06	3.9751E+06	2.1843E+06	5.0223E+05	8.1290E+05	2.3729E+05
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F15	Best	4.3232E+04	4.8546E+06	3.2940E+05	7.5029E+05	2.8137E+04	3.3688E+04	1.6071E+04	3.1641E+03	2.1018E+03
	Worst	6.1167E+08	6.7997E+08	1.6667E+08	3.3191E+07	1.8735E+09	4.5660E+09	1.4732E+05	3.2676E+04	1.9803E+04
	Ave	3.8641E+07	7.3678E+07	8.1314E+06	8.6006E+06	8.0247E+07	2.5434E+08	5.3564E+04	1.4319E+04	6.5176E+03
	Std	1.1125E+08	1.2457E+08	3.0214E+07	6.4811E+06	3.4218E+08	8.7701E+08	4.0115E+04	6.8698E+03	4.8541E+03
F16	Best	2.4500E+03	4.8897E+03	3.6950E+03	3.2061E+03	3.3626E+03	3.1401E+03	3.7431E+03	2.4714E+03	2.3788E+03
	Worst	4.4377E+03	7.9175E+03	7.0047E+03	5.3587E+03	6.1235E+03	9.2461E+03	6.4254E+03	4.2538E+03	3.8827E+03
	Ave	3.4412E+03	6.5963E+03	4.8828E+03	4.2522E+03	4.6986E+03	4.5869E+03	5.0518E+03	3.2873E+03	3.1371E+03
	Std	4.3637E+02	9.1313E+02	7.3917E+02	4.8055E+02	6.7571E+02	1.2220E+03	7.1670E+02	3.8012E+02	3.8745E+02
F17	Best	2.4142E+03	3.4126E+03	2.8966E+03	3.1379E+03	3.0772E+03	2.8724E+03	3.3659E+03	2.3647E+03	2.4930E+03
	Worst	3.8521E+03	6.1016E+03	4.4988E+03	4.4462E+03	5.4772E+03	6.0316E+03	5.0450E+03	3.6318E+03	3.9678E+03
	Ave	3.0415E+03	4.5791E+03	3.8891E+03	3.8209E+03	4.0920E+03	3.7239E+03	3.9406E+03	3.1140E+03	3.0886E+03
	Std	3.0401E+02	5.9110E+02	3.8078E+02	2.9783E+02	6.3122E+02	6.3950E+02	3.9971E+02	3.6708E+02	3.4740E+02
F18	Best	1.0159E+06	8.5908E+06	1.5550E+06	1.7599E+06	9.7288E+05	4.1910E+05	2.7154E+05	6.7027E+05	3.0641E+05
	Worst	5.9101E+07	1.5328E+08	4.9335E+07	1.8467E+07	4.2763E+07	9.5204E+06	1.3980E+07	1.5970E+07	6.5253E+06
	Ave	1.0319E+07	7.0304E+07	1.3035E+07	7.3082E+06	1.1249E+07	1.9550E+06	2.8012E+06	3.7041E+06	2.2234E+06
	Std	1.4577E+07	3.6747E+07	1.0772E+07	4.7263E+06	1.0220E+07	2.0012E+06	2.9939E+06	3.5259E+06	1.7525E+06
F19	Best	6.7922E+04	2.5159E+06	2.6136E+05	1.3165E+06	6.6381E+04	5.6948E+04	7.7466E+04	2.1854E+03	2.1962E+03
	Worst	1.2964E+08	8.0081E+07	1.0017E+07	1.6393E+07	3.6079E+07	4.9809E+09	7.2701E+06	4.4251E+04	3.3252E+04
	Ave	1.2817E+07	1.7788E+07	2.4128E+06	5.7528E+06	8.2759E+06	2.4660E+08	2.6596E+06	2.0382E+04	1.3923E+04
	Std	2.7786E+07	1.7919E+07	2.3660E+06	4.0798E+06	1.1369E+07	9.3980E+08	2.2124E+06	1.2152E+04	8.4448E+03
F20	Best	2.6576E+03	3.2668E+03	3.0748E+03	3.0168E+03	3.0051E+03	2.7606E+03	3.2793E+03	2.3017E+03	2.5224E+03
	Worst	3.8857E+03	4.7556E+03	4.1624E+03	4.0554E+03	4.6628E+03	4.4517E+03	4.7918E+03	3.6982E+03	3.5310E+03
	Ave	3.1895E+03	4.0150E+03	3.6433E+03	3.5958E+03	3.8287E+03	3.3294E+03	4.0406E+03	3.0965E+03	3.0279E+03
	Std	2.9739E+02	3.9212E+02	2.8750E+02	2.5055E+02	3.9839E+02	4.0479E+02	3.9489E+02	3.4885E+02	3.2515E+02
F21	Best	2.4696E+03	2.9192E+03	2.7531E+03	2.6480E+03	2.6761E+03	2.6478E+03	2.9161E+03	2.4720E+03	2.4170E+03
	Worst	2.6497E+03	3.3368E+03	3.0615E+03	2.8866E+03	3.0102E+03	3.1408E+03	3.2811E+03	2.6507E+03	2.5422E+03
	Ave	2.5449E+03	3.0921E+03	2.9312E+03	2.7835E+03	2.8443E+03	2.8682E+03	3.1042E+03	2.5394E+03	2.4765E+03
	Std	4.9395E+01	9.9672E+01	7.9580E+01	4.8095E+01	7.7126E+01	1.1284E+02	8.9959E+01	4.7222E+01	3.0402E+01
F22	Best	8.1584E+03	1.2877E+04	1.0008E+04	3.6974E+03	8.8790E+03	8.4463E+03	8.9842E+03	2.8223E+03	8.3224E+03
	Worst	1.6753E+04	1.6797E+04	1.4280E+04	1.6542E+04	1.7123E+04	1.6329E+04	1.3095E+04	1.1740E+04	1.0739E+04
	Ave	1.0765E+04	1.4816E+04	1.2415E+04	1.4141E+04	1.3744E+04	1.1321E+04	1.0896E+04	9.1820E+03	9.5518E+03
	Std	2.4869E+03	1.0440E+03	9.2905E+02	2.9892E+03	2.4577E+03	1.6279E+03	1.0157E+03	2.3846E+03	7.2546E+02
F23	Best	2.9117E+03	3.4081E+03	3.6987E+03	3.2637E+03	3.3323E+03	3.4130E+03	3.8071E+03	2.9314E+03	2.9311E+03
	Worst	3.2210E+03	4.0777E+03	4.5076E+03	3.6322E+03	3.8278E+03	4.4521E+03	4.7582E+03	3.2768E+03	3.2106E+03
	Ave	3.0512E+03	3.7909E+03	4.0430E+03	3.4490E+03	3.5454E+03	3.7721E+03	4.2615E+03	3.1304E+03	3.0484E+03
	Std	7.9240E+01	1.5105E+02	2.3937E+02	1.0686E+02	1.3692E+02	2.0794E+02	2.4212E+02	7.4404E+01	7.2886E+01
F24	Best	3.1396E+03	3.6383E+03	3.9087E+03	3.4034E+03	3.4523E+03	3.4185E+03	4.1384E+03	3.1623E+03	3.0862E+03
	Worst	3.5514E+03	4.3490E+03	5.0197E+03	4.0486E+03	4.0649E+03	4.2195E+03	4.6586E+03	3.4701E+03	3.7857E+03
	Ave	3.2635E+03	3.9110E+03	4.3664E+03	3.6797E+03	3.7731E+03	3.7910E+03	4.4375E+03	3.2889E+03	3.4048E+03
	Std	1.1167E+02	1.6633E+02	2.7696E+02	1.6157E+02	1.6323E+02	1.7131E+02	1.2754E+02	7.9421E+01	1.7180E+02
F25	Best	3.2455E+03	4.4857E+03	3.4650E+03	3.2406E+03	3.2147E+03	4.2756E+03	3.2574E+03	3.2739E+03	3.1350E+03
	Worst	4.7165E+03	6.5542E+03	4.5722E+03	3.6099E+03	1.2927E+04	1.4497E+04	7.9654E+03	3.9069E+03	3.3497E+03
	Ave	3.9094E+03	5.3514E+03	3.7612E+03	3.3829E+03	4.0777E+03	6.1002E+03	4.2418E+03	3.4793E+03	3.2220E+03
	Std	3.7513E+02	5.0884E+02	2.3008E+02	8.2246E+01	1.9466E+03	2.1678E+03	1.1740E+03	1.4099E+02	4.9582E+01
F26	Best	5.9474E+03	1.0913E+04	7.4199E+03	4.1720E+03	4.6489E+03	8.0941E+03	7.6700E+03	5.1732E+03	6.6734E+03
	Worst	1.0108E+04	1.7825E+04	1.3482E+04	1.0154E+04	1.3710E+04	1.8979E+04	2.2206E+04	1.1398E+04	9.9896E+03
	Ave	7.1132E+03	1.5306E+04	1.1750E+04	8.0477E+03	1.0757E+04	1.2773E+04	1.5544E+04	8.7287E+03	7.8912E+03
	Std	7.9838E+02	1.5266E+03	1.2872E+03	1.5572E+03	1.9891E+03	2.0261E+03	4.8786E+03	1.6229E+03	8.7777E+02
F27	Best	3.5636E+03	3.9266E+03	4.1116E+03	3.4130E+03	3.6185E+03	3.6848E+03	4.1839E+03	3.6038E+03	3.2000E+03
	Worst	4.1621E+03	5.9385E+03	6.2872E+03	4.5587E+03	4.4201E+03	5.2144E+03	7.3644E+03	3.9890E+03	3.9245E+03
	Ave	3.7165E+03	4.8740E+03	5.1590E+03	3.7414E+03	3.9985E+03	4.3162E+03	5.9676E+03	3.8064E+03	3.3590E+03
	Std	1.2305E+02	5.8664E+02	6.9587E+02	2.7940E+02	2.3085E+02	4.0338E+02	8.3840E+02	1.0564E+02	2.4671E+02
F28	Best	3.8144E+03	4.9801E+03	4.0372E+03	3.4319E+03	3.5023E+03	4.3313E+03	4.0828E+03	3.7841E+03	3.3002E+03
	Worst	5.2331E+03	6.9573E+03	5.6796E+03	5.0727E+03	1.0815E+04	1.2382E+04	9.0862E+03	5.0433E+03	3.9279E+03
	Ave	4.5135E+03	5.9311E+03	4.8920E+03	3.9714E+03	6.4913E+03	6.7969E+03	5.6703E+03	4.2614E+03	3.6254E+03
	Std	3.8367E+02	5.7607E+02	3.5488E+02	4.6374E+02	2.3862E+03	2.4748E+03	1.2555E+03	2.9802E+02	1.2959E+02
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F29	Best	4.1921E+03	5.7256E+03	5.5488E+03	4.8617E+03	4.5668E+03	4.8079E+03	5.5266E+03	4.0582E+03	3.9723E+03
	Worst	5.8342E+03	1.4555E+04	9.9456E+03	6.3185E+03	1.0951E+04	5.1878E+04	1.1421E+04	5.3393E+03	5.2544E+03
	Ave	4.9409E+03	9.7863E+03	7.3002E+03	5.6285E+03	6.4811E+03	9.7573E+03	7.2440E+03	4.6230E+03	4.5860E+03
	Std	3.8479E+02	2.0660E+03	1.1051E+03	4.3515E+02	1.2686E+03	8.9407E+03	1.1060E+03	3.7819E+02	2.7130E+02
F30	Best	7.4623E+07	1.1891E+08	6.1237E+07	4.7145E+07	5.7089E+06	3.1661E+07	5.6145E+07	4.4654E+06	1.8902E+05
	Worst	2.7740E+08	5.6622E+08	3.3859E+08	2.0795E+08	1.3257E+08	4.2865E+09	3.8769E+08	2.3576E+07	4.3650E+07
	Ave	1.6038E+08	2.7800E+08	1.1987E+08	1.0267E+08	4.3327E+07	4.9198E+08	1.1130E+08	1.0188E+07	9.1542E+06
	Std	4.3556E+07	1.0895E+08	6.3043E+07	3.6862E+07	3.6523E+07	1.1557E+09	6.9204E+07	4.0680E+06	8.5317E+06
Mean		4.10	8.57	6.30	4.80	5.73	5.30	6.27	2.67	1.27
Ranking		3	9	8	4	6	5	7	2	1

Table 5. Experimental evaluation outcomes utilizing CEC2017 test functions (dim = 50).

Where the coefficient C stands for the penalty parameter, and I denotes the identity matrix. Thus, the expression for the output function can be defined as:

$$f(x) = h(x) H^T \left(\frac{I}{C} + H H^T \right)^{-1} T \quad (28)$$

The kernel matrix of ELM can be obtained as follows:

$$\Omega_{ELM} = H H^T : \Omega_{ELM_{i,j}} = h(x_i)^2 = K(x_i, x_i) \quad (29)$$

Where $K(x_i, x_i)$ represents a kernel function. The output function can be expressed as:

$$f(x) = \begin{bmatrix} K(x, x_1) \\ \vdots \\ K(x, x_N) \end{bmatrix}^T \times \left(\frac{I}{C} + \Omega_{ELM} \right)^{-1} T \quad (30)$$

KELM, or the ELM implemented with kernel-based methods, brings higher stability and generalization performance to ELM through its kernelization. Figure 14 briefly illustrates the structure of the KELM model, where the kernel function replaces the traditional feature mapping function, achieving the transformation from the input to the feature space. This means that the output of the neural network no longer depends on the feature mapping of the hidden layer but is explicitly determined by the kernel function. Additionally, the Dimensionality of the feature mapping or the feature space is no longer predefined.

In this research, we employ the Gaussian radial basis function as the kernel function for KELM, described by the subsequent equation:

$$K(u, v) = \exp(-\gamma \|u - v\|^2) \quad (31)$$

The penalty parameter C and the kernel parameter c are pivotal in shaping the model. C strikes a balance between minimizing fitting errors and managing model complexity. Meanwhile, c specifies the non-linear transformation from the input space to a particular high-Dimensional hidden feature space. Generally, optimizing the performance of KELM involves selecting these key parameters using appropriate optimization techniques.

KELM improved by MEARO

In this section, we introduce an enhanced version of the KELM model called MEARO-KELM. This model adopts the MEARO optimization algorithm mentioned earlier in the text, which can adaptively determine the two key parameters of KELM: the penalty parameter C and the kernel parameter c . The range of optimization parameters set in this paper is shown in Table 9. The model consists mainly of two procedures. The initial procedure focuses on optimizing the parameters in the inner loop, whereas the second procedure assesses the classification attributes in the outer loop. Throughout the parameter optimization, MEARO dynamically tunes the KELM parameters. Once we have the optimal parameters, they're integrated into the KELM classification model to address the classification challenge. The fitness function f is crafted with classification accuracy in mind:

$$f = avgAcc = \sum_i^K \frac{testAcc_i}{K} \quad (32)$$

Here $avgAcc$ stands for the average test accuracy achieved by the KELM classifier using 5-fold cross-validation during the parameter tuning phase.

The framework for score prediction using the MEARO-KELM approach is depicted in Fig. 15. Consistent with many studies, we employ 10-fold cross-validation (CV) in the proposed predictive system to assess the classifier's performance, aiming for reliable and unbiased results in this case study. Internally, within each

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F1	Best	3.7413E+10	8.7399E+10	3.4547E+10	1.8467E+10	2.2336E+10	1.1914E+11	1.9953E+10	3.0602E+10	2.1647E+09
	Worst	9.1689E+10	1.3695E+11	6.9165E+10	6.2741E+10	2.6008E+11	2.6476E+11	1.6234E+11	6.7129E+10	1.1282E+10
	Ave	5.7528E+10	1.0957E+11	5.0755E+10	3.1940E+10	1.0049E+11	1.4371E+11	7.3866E+10	5.1101E+10	4.8977E+09
	Std	1.0008E+10	1.1109E+10	9.1334E+09	1.0420E+10	7.7606E+10	2.6204E+10	5.6933E+10	9.6080E+09	2.2899E+09
F2	Best	1.4764E+106	2.9077E+147	1.7463E+134	7.7830E+116	1.7632E+130	2.1359E+121	4.7625E+122	2.0652E+116	2.3254E+102
	Worst	9.6563E+134	1.8308E+178	7.7439E+154	1.5926E+138	6.0203E+153	2.1107E+175	1.0838E+158	7.3959E+141	2.1120E+122
	Ave	3.4363E+133	6.1780E+176	2.7419E+153	5.3086E+136	3.4492E+152	7.0771E+173	3.6203E+156	2.4654E+140	7.2449E+120
	Std	1.7606E+134	6.5535E+04	6.5535E+04	2.9076E+137	1.2978E+153	6.5535E+04	6.5535E+04	1.3503E+141	3.8526E+121
F3	Best	4.6676E+05	5.7451E+05	3.1533E+05	4.2504E+05	3.5996E+05	2.0369E+05	6.0742E+05	3.3505E+05	2.0184E+05
	Worst	7.0375E+05	1.5728E+06	5.3420E+05	7.3992E+05	1.0168E+06	3.3476E+05	1.0486E+06	4.5824E+05	3.7029E+05
	Ave	5.7214E+05	9.2172E+05	3.5058E+05	5.6165E+05	6.1783E+05	2.6251E+05	8.4725E+05	3.8697E+05	2.6350E+05
	Std	6.5995E+04	1.8500E+05	3.6331E+04	8.4825E+04	2.2044E+05	2.4364E+04	1.0171E+05	3.3416E+04	3.3158E+04
F4	Best	3.8938E+03	1.3150E+04	5.7214E+03	2.0777E+03	2.5849E+03	9.7274E+03	3.5225E+03	3.7011E+03	1.2851E+03
	Worst	1.4620E+04	3.0837E+04	1.4130E+04	7.5513E+03	5.0576E+04	9.9560E+04	2.6295E+04	7.6039E+03	2.4286E+03
	Ave	6.4188E+03	2.1634E+04	9.1109E+03	3.8626E+03	1.2952E+04	2.9208E+04	1.0849E+04	5.5589E+03	1.8411E+03
	Std	2.1915E+03	4.6066E+03	1.8863E+03	1.4567E+03	1.2504E+04	2.3483E+04	8.1640E+03	9.2071E+02	2.7383E+02
F5	Best	1.0784E+03	1.7899E+03	1.5492E+03	1.4986E+03	1.3584E+03	1.3675E+03	1.3214E+03	1.2433E+03	9.6248E+02
	Worst	1.3675E+03	2.3932E+03	1.7687E+03	1.8982E+03	2.1036E+03	2.0941E+03	2.4837E+03	1.6047E+03	1.2034E+03
	Ave	1.2275E+03	1.9731E+03	1.6744E+03	1.7056E+03	1.7498E+03	1.5737E+03	1.9531E+03	1.4248E+03	1.0776E+03
	Std	6.6756E+01	1.3434E+02	5.3603E+01	9.4532E+01	2.4803E+02	1.8428E+02	4.2698E+02	8.0611E+01	6.6903E+01
F6	Best	6.3360E+02	6.9121E+02	6.8326E+02	6.4941E+02	6.5309E+02	6.7110E+02	6.6561E+02	6.4396E+02	6.3222E+02
	Worst	6.5853E+02	7.3214E+02	6.9881E+02	6.9962E+02	7.1195E+02	7.0928E+02	7.0035E+02	6.6718E+02	6.5176E+02
	Ave	6.4735E+02	7.0840E+02	6.9191E+02	6.7257E+02	6.8217E+02	6.8274E+02	6.8477E+02	6.5854E+02	6.4377E+02
	Std	5.1839E+00	9.6432E+00	3.8615E+00	1.1443E+01	1.5777E+01	9.6654E+00	1.1909E+01	5.9058E+00	5.0880E+00
F7	Best	1.9284E+03	3.3908E+03	3.5339E+03	2.2203E+03	2.3748E+03	3.1913E+03	3.2686E+03	2.5287E+03	1.6780E+03
	Worst	2.5374E+03	4.4639E+03	3.9724E+03	3.1454E+03	3.3012E+03	4.0595E+03	1.1793E+04	3.3345E+03	2.5057E+03
	Ave	2.1865E+03	3.8311E+03	3.7757E+03	2.4686E+03	2.8838E+03	3.4377E+03	8.1691E+03	2.9473E+03	2.1258E+03
	Std	1.5228E+02	2.0923E+02	1.0949E+02	1.7937E+02	2.6775E+02	2.2154E+02	3.4653E+03	2.2345E+02	1.7741E+02
F8	Best	1.4265E+03	2.2442E+03	2.0377E+03	1.8617E+03	1.7434E+03	1.7875E+03	1.8464E+03	1.6432E+03	1.2853E+03
	Worst	1.7109E+03	2.7972E+03	2.2576E+03	2.1885E+03	2.5648E+03	2.0789E+03	2.8009E+03	1.9560E+03	1.5836E+03
	Ave	1.5374E+03	2.4708E+03	2.1387E+03	2.0212E+03	2.1977E+03	1.9397E+03	2.2185E+03	1.7737E+03	1.4519E+03
	Std	7.1942E+01	1.4519E+02	5.7013E+01	9.4880E+01	2.2518E+02	8.2135E+01	3.4829E+02	7.6473E+01	6.4836E+01
F9	Best	2.6503E+04	5.7391E+04	5.8370E+04	2.5140E+04	4.5356E+04	3.0369E+04	2.8005E+04	2.8470E+04	1.2467E+04
	Worst	7.2175E+04	1.2385E+05	7.7188E+04	1.0116E+05	8.9735E+04	7.7663E+04	7.7995E+04	4.5167E+04	2.7541E+04
	Ave	4.3791E+04	7.8398E+04	6.8641E+04	6.4422E+04	7.6271E+04	3.7241E+04	4.3283E+04	3.3431E+04	2.0256E+04
	Std	1.3827E+04	1.7046E+04	4.7040E+03	1.8980E+04	1.1550E+04	1.0659E+04	1.7290E+04	3.2890E+03	3.4970E+03
F10	Best	1.6311E+04	2.6425E+04	2.1796E+04	2.7778E+04	1.9256E+04	1.7113E+04	1.3906E+04	1.7110E+04	1.3684E+04
	Worst	3.3189E+04	3.2259E+04	2.7871E+04	3.1694E+04	3.3925E+04	3.2154E+04	1.9334E+04	2.1857E+04	2.0680E+04
	Ave	1.9644E+04	2.9520E+04	2.4478E+04	2.9720E+04	2.8368E+04	2.1543E+04	1.7161E+04	1.9762E+04	1.7126E+04
	Std	4.5325E+03	1.5014E+03	1.4277E+03	1.1523E+03	4.6619E+03	4.2399E+03	1.1968E+03	1.2894E+03	1.5411E+03
F11	Best	6.9501E+04	1.3764E+05	7.5492E+04	5.6982E+04	1.3951E+05	3.2835E+04	1.2510E+05	4.5225E+04	1.6455E+04
	Worst	1.3885E+05	6.0007E+05	2.2546E+05	1.2635E+05	3.4481E+05	1.5469E+05	3.2127E+05	1.2875E+05	4.8952E+04
	Ave	9.3637E+04	3.1774E+05	1.4588E+05	8.9946E+04	2.2991E+05	7.2662E+04	1.9414E+05	8.6012E+04	3.1278E+04
	Std	1.7821E+04	1.0715E+05	3.3709E+04	1.9041E+04	4.6235E+04	2.7968E+04	4.5457E+04	1.9683E+04	7.7098E+03
F12	Best	4.4731E+09	1.8930E+10	4.5155E+09	3.6037E+09	3.6879E+09	2.6521E+10	1.0529E+09	1.8631E+09	2.2724E+08
	Worst	2.7423E+10	4.6899E+10	1.8827E+10	3.8064E+10	1.3730E+10	1.9044E+11	3.2927E+10	9.8085E+09	1.0920E+09
	Ave	1.1502E+10	2.9757E+10	1.1041E+10	1.1591E+10	7.9276E+09	6.2743E+10	1.3623E+10	3.4536E+09	4.5447E+08
	Std	5.0374E+09	7.9147E+09	3.6567E+09	8.4725E+09	2.3071E+09	3.6899E+10	1.0963E+10	1.7620E+09	1.7767E+08
F13	Best	6.3948E+06	1.0875E+09	1.0691E+08	2.9168E+08	2.6441E+07	1.4970E+09	3.3697E+04	6.1333E+06	2.7437E+04
	Worst	3.5976E+09	5.7147E+09	7.5857E+08	3.4439E+09	8.0754E+08	4.8007E+10	5.1979E+08	4.1181E+07	3.4034E+06
	Ave	1.2666E+09	2.8009E+09	2.9363E+08	1.3808E+09	3.1347E+08	8.7378E+09	1.8919E+07	1.8673E+07	2.5997E+05
	Std	1.0103E+09	1.1816E+09	1.7166E+08	9.2121E+08	2.2268E+08	9.2016E+09	9.4967E+07	1.0271E+07	6.3346E+05
F14	Best	2.3030E+06	9.1938E+06	5.5332E+06	4.1439E+06	1.7634E+06	4.1155E+05	1.0415E+06	1.4314E+06	7.6492E+05
	Worst	1.8756E+07	3.2454E+07	1.5426E+07	3.5581E+07	7.5509E+07	4.9653E+06	1.7449E+07	1.0739E+07	5.7203E+06
	Ave	8.6923E+06	2.0193E+07	1.0133E+07	1.4993E+07	2.2837E+07	1.7129E+06	5.3459E+06	5.2309E+06	2.9363E+06
	Std	4.3576E+06	6.4066E+06	2.7630E+06	8.3964E+06	1.7724E+07	1.0126E+06	3.3989E+06	2.1931E+06	1.2980E+06
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F15	Best	1.8658E+07	1.3300E+08	5.7653E+06	3.7581E+07	1.1311E+05	8.8412E+05	2.2472E+04	5.6710E+04	4.5098E+03
	Worst	1.3299E+09	1.4735E+09	3.6375E+07	1.3664E+09	3.2670E+08	2.5046E+10	1.9608E+05	1.0139E+06	7.0803E+04
	Ave	2.4127E+08	5.0063E+08	1.4780E+07	2.5195E+08	7.3090E+07	3.1204E+09	6.1562E+04	2.3723E+05	1.5171E+04
	Std	2.7071E+08	3.1275E+08	7.6765E+06	3.3196E+08	9.1446E+07	6.8341E+09	4.5960E+04	1.9124E+05	1.4587E+04
F16	Best	5.4921E+03	1.2265E+04	7.5215E+03	8.0799E+03	7.3123E+03	7.1521E+03	6.5843E+03	5.7434E+03	5.0762E+03
	Worst	8.2645E+03	2.1652E+04	1.3787E+04	1.1971E+04	1.4341E+04	2.5627E+04	1.1281E+04	7.8549E+03	7.3934E+03
	Ave	6.8726E+03	1.7025E+04	1.0483E+04	9.7478E+03	9.6485E+03	1.0644E+04	8.7671E+03	6.7818E+03	5.9556E+03
	Std	7.3185E+02	2.3842E+03	1.3350E+03	9.0220E+02	1.8929E+03	3.9744E+03	1.2674E+03	5.6505E+02	6.1853E+02
F17	Best	4.1227E+03	8.7678E+03	5.9832E+03	6.8837E+03	7.4483E+03	5.8997E+03	5.1451E+03	4.2347E+03	3.8745E+03
	Worst	1.0572E+04	9.3778E+04	1.4276E+04	2.3625E+04	1.3007E+04	1.4876E+05	7.7312E+03	6.7311E+03	5.8848E+03
	Ave	5.8001E+03	2.7539E+04	8.2397E+03	8.9720E+03	1.0109E+04	1.6594E+04	6.5634E+03	5.3766E+03	4.9142E+03
	Std	1.1489E+03	2.2463E+04	1.4862E+03	3.5164E+03	1.4722E+03	2.8662E+04	6.5737E+02	5.5964E+02	4.6868E+02
F18	Best	2.9112E+06	6.2306E+06	3.1856E+06	4.9088E+06	5.4672E+06	9.5309E+05	1.0417E+06	2.6776E+06	7.2471E+05
	Worst	4.2937E+07	3.7869E+07	2.0433E+07	3.6057E+07	1.0319E+08	9.0376E+07	1.0810E+07	1.0014E+07	5.9143E+06
	Ave	1.1082E+07	1.8559E+07	9.8543E+06	1.6860E+07	2.5694E+07	1.2984E+07	3.7619E+06	5.8550E+06	3.1173E+06
	Std	8.3904E+06	8.9891E+06	4.7572E+06	7.4834E+06	2.0297E+07	2.5392E+07	2.4609E+06	2.1621E+06	1.1829E+06
F19	Best	7.3886E+06	1.2676E+08	1.4254E+07	1.0969E+08	1.2076E+07	1.6801E+07	9.4699E+05	2.2827E+05	3.7189E+03
	Worst	1.9431E+09	1.0038E+09	3.4900E+08	8.7335E+08	2.6205E+08	1.8831E+10	3.1763E+07	1.2455E+06	9.5465E+04
	Ave	2.2818E+08	5.3802E+08	4.4233E+07	3.4438E+08	9.8927E+07	3.1590E+09	1.3029E+07	6.5441E+05	2.3983E+04
	Std	3.5229E+08	2.6458E+08	5.9439E+07	1.9261E+08	7.0302E+07	5.7796E+09	1.0790E+07	2.7495E+05	2.5410E+04
F20	Best	4.2618E+03	6.0789E+03	4.9865E+03	6.2326E+03	6.1457E+03	4.6089E+03	5.1648E+03	4.1090E+03	4.1350E+03
	Worst	8.0256E+03	8.6551E+03	7.1070E+03	8.1873E+03	8.3173E+03	7.7863E+03	7.6727E+03	6.1684E+03	5.9171E+03
	Ave	5.6816E+03	7.1952E+03	6.0901E+03	7.0572E+03	7.5247E+03	5.7361E+03	6.5888E+03	5.1824E+03	5.0630E+03
	Std	1.1040E+03	6.6379E+02	6.1752E+02	4.5102E+02	5.8872E+02	7.5102E+02	5.9425E+02	5.4618E+02	4.2299E+02
F21	Best	2.9429E+03	3.9863E+03	4.0919E+03	3.4943E+03	3.6335E+03	3.8482E+03	3.9411E+03	2.9696E+03	2.7950E+03
	Worst	3.2906E+03	4.9743E+03	4.8000E+03	4.0383E+03	4.3415E+03	4.7527E+03	4.7563E+03	3.4409E+03	3.1046E+03
	Ave	3.1065E+03	4.4470E+03	4.4026E+03	3.7166E+03	4.0122E+03	4.1009E+03	4.4236E+03	3.2087E+03	2.9397E+03
	Std	7.8007E+01	2.2900E+02	1.7024E+02	1.4212E+02	1.6092E+02	2.1612E+02	1.6406E+02	1.1015E+02	7.8985E+01
F22	Best	1.8702E+04	2.6401E+04	2.5341E+04	2.9478E+04	2.1172E+04	2.0429E+04	1.7608E+04	1.9553E+04	1.6824E+04
	Worst	3.5592E+04	3.4751E+04	3.0971E+04	3.4650E+04	3.6315E+04	3.4975E+04	2.3426E+04	2.5093E+04	2.3493E+04
	Ave	2.1829E+04	3.1819E+04	2.7299E+04	3.2129E+04	2.9223E+04	2.4639E+04	2.0062E+04	2.2630E+04	2.0037E+04
	Std	4.5441E+03	1.8790E+03	1.3823E+03	1.3682E+03	4.8353E+03	3.6378E+03	1.7380E+03	1.4914E+03	1.6146E+03
F23	Best	3.5194E+03	4.8870E+03	5.2883E+03	4.3544E+03	4.2273E+03	4.5841E+03	5.2779E+03	3.6791E+03	3.7008E+03
	Worst	3.9029E+03	5.8209E+03	6.8042E+03	5.3604E+03	5.2072E+03	6.2357E+03	6.7230E+03	4.2452E+03	4.9385E+03
	Ave	3.6871E+03	5.3213E+03	5.9195E+03	4.8621E+03	4.8142E+03	5.1773E+03	5.9521E+03	3.8577E+03	4.3963E+03
	Std	8.1961E+01	2.6335E+02	3.5674E+02	2.6622E+02	2.2547E+02	3.1898E+02	3.4998E+02	1.3354E+02	2.2027E+02
F24	Best	4.1590E+03	6.1956E+03	6.8096E+03	5.2532E+03	5.4109E+03	6.0139E+03	7.1847E+03	4.3871E+03	4.6644E+03
	Worst	5.0773E+03	7.5756E+03	9.4971E+03	6.7492E+03	6.7526E+03	8.8751E+03	8.6577E+03	5.1834E+03	6.9484E+03
	Ave	4.4949E+03	6.7160E+03	8.3994E+03	5.7519E+03	6.1461E+03	6.8121E+03	7.8658E+03	4.8647E+03	5.8787E+03
	Std	1.8813E+02	3.7261E+02	6.2803E+02	3.5016E+02	3.5421E+02	5.7410E+02	3.6775E+02	2.1923E+02	6.3737E+02
F25	Best	5.5579E+03	8.4736E+03	5.5179E+03	4.6381E+03	5.2022E+03	8.0778E+03	4.4647E+03	5.8671E+03	4.0031E+03
	Worst	9.7119E+03	1.3275E+04	8.1157E+03	8.4511E+03	2.0701E+04	2.7036E+04	1.7592E+04	8.3033E+03	4.7427E+03
	Ave	7.1534E+03	1.0865E+04	6.8577E+03	5.7177E+03	7.5529E+03	1.3501E+04	7.7682E+03	6.7699E+03	4.3281E+03
	Std	1.0015E+03	1.0255E+03	5.9648E+02	7.5218E+02	3.5308E+03	4.4893E+03	4.2767E+03	6.6011E+02	1.9088E+02
F26	Best	1.5350E+04	3.3513E+04	2.7833E+04	1.6363E+04	1.9956E+04	3.0874E+04	2.2075E+04	2.1700E+04	1.7533E+04
	Worst	2.2546E+04	4.8168E+04	3.7730E+04	2.8468E+04	3.2247E+04	5.8548E+04	4.8746E+04	3.0698E+04	2.4891E+04
	Ave	1.7747E+04	3.9806E+04	3.2161E+04	2.1280E+04	2.5693E+04	3.7814E+04	3.3474E+04	2.6169E+04	2.1413E+04
	Std	1.6126E+03	3.4493E+03	2.1476E+03	2.6600E+03	3.5818E+03	6.7139E+03	9.3689E+03	2.7783E+03	1.7876E+03
F27	Best	3.9266E+03	4.8997E+03	5.8804E+03	3.6407E+03	3.9255E+03	4.9394E+03	4.8792E+03	4.1139E+03	3.2000E+03
	Worst	5.2771E+03	8.9355E+03	1.3614E+04	4.7353E+03	5.8062E+03	9.0144E+03	1.3109E+04	5.1749E+03	6.5766E+03
	Ave	4.4080E+03	6.5869E+03	7.4990E+03	3.9549E+03	4.4427E+03	6.1791E+03	8.0225E+03	4.5288E+03	4.1440E+03
	Std	2.7286E+02	9.3862E+02	1.5229E+03	2.1570E+02	3.8435E+02	1.0201E+03	1.9328E+03	2.5325E+02	8.0630E+02
F28	Best	6.1604E+03	1.2029E+04	7.6627E+03	4.5280E+03	8.5070E+03	1.2316E+04	8.0739E+03	7.0477E+03	4.1524E+03
	Worst	1.4975E+04	1.7837E+04	1.1888E+04	1.0340E+04	3.0101E+04	3.2806E+04	2.7844E+04	1.1964E+04	6.3274E+03
	Ave	9.8124E+03	1.4703E+04	9.5536E+03	7.1598E+03	2.0994E+04	1.7698E+04	1.6472E+04	9.2809E+03	5.4175E+03
	Std	1.6918E+03	1.5416E+03	1.0469E+03	1.8081E+03	5.6810E+03	4.4240E+03	5.5318E+03	1.0494E+03	5.0017E+02
Continued										

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F29	Best	7.7005E+03	1.5586E+04	1.1198E+04	9.4329E+03	9.4114E+03	9.8576E+03	9.6592E+03	7.6242E+03	6.3445E+03
	Worst	1.0833E+04	3.2423E+04	1.5653E+04	1.6663E+04	1.8970E+04	1.8951E+05	1.8019E+04	1.0022E+04	9.5756E+03
	Ave	9.3135E+03	2.1374E+04	1.3387E+04	1.1008E+04	1.1632E+04	2.1954E+04	1.2524E+04	8.8417E+03	8.0204E+03
	Std	7.1983E+02	4.0272E+03	1.0763E+03	1.3053E+03	2.0998E+03	3.3835E+04	2.1555E+03	6.1206E+02	7.3249E+02
F30	Best	1.7575E+08	9.7965E+08	2.2253E+08	2.2532E+08	5.9147E+07	3.3741E+08	5.8425E+07	1.5523E+07	1.3878E+06
	Worst	4.6065E+09	5.5469E+09	1.9832E+09	4.1803E+09	1.0850E+09	4.0464E+10	1.1597E+09	1.4438E+08	1.9347E+08
	Ave	1.4513E+09	2.6905E+09	7.3146E+08	1.2948E+09	2.9835E+08	8.3039E+09	3.8183E+08	6.7528E+07	2.1562E+07
	Std	1.1026E+09	1.0581E+09	3.7560E+08	1.0590E+09	1.9103E+08	1.0487E+10	3.0124E+08	3.4635E+07	3.7326E+07
Mean		3.67	8.37	6.13	4.87	6.07	6.03	5.43	3.13	1.30
Ranking		3	9	8	4	7	6	5	2	1

Table 6. Experimental evaluation outcomes utilizing CEC2017 test functions (dim = 100).

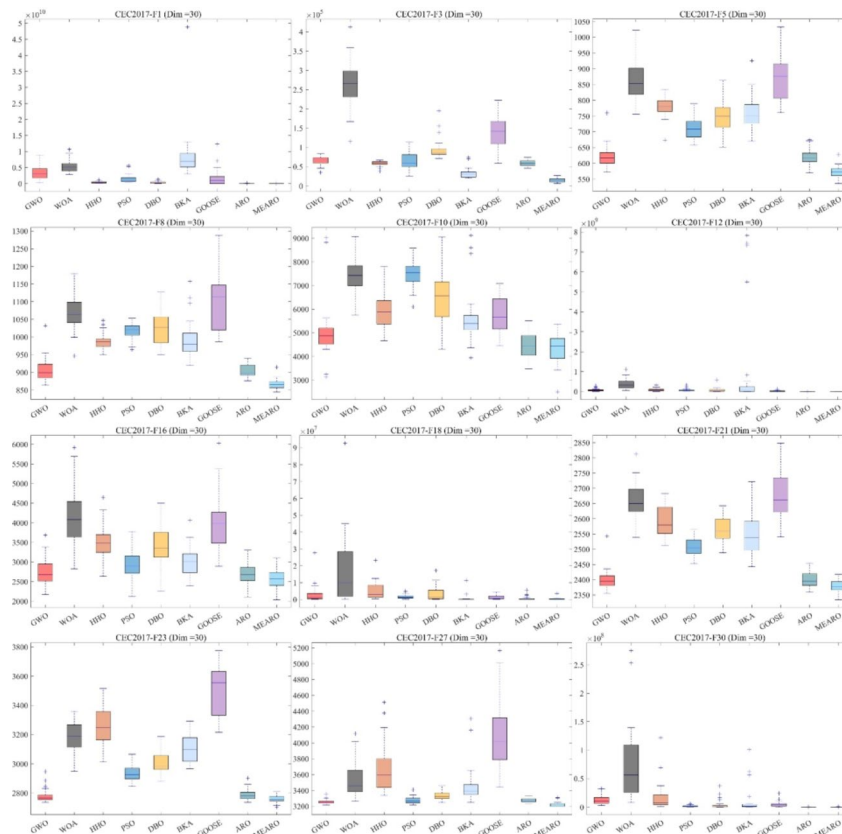


Fig. 8. Boxplot of different algorithms in optimization of the CEC-2017 (Dim = 30).

experiment, 5-fold CV is utilized to fine-tune the two classifier parameters. In each experiment iteration based on the 10-fold CV, nine subsets are chosen to constitute a specific sample, with the remaining subset acting as the test set. Moreover, the fifth subset from this specific sample is designated as the validation set, while the others serve as the training set. This approach aids in obtaining unbiased generalization accuracy estimates and dependable results. It's important to mention that we use a stratified strategy to partition the dataset. This means that each fold's samples are chosen according to the ratio of good to bad scores in the complete dataset. Due to the randomness inherent in sampling, a single 10-fold CV run might not yield satisfactory classification accuracy. Hence, for each method under consideration, we conduct 30 runs of 10-fold CV, averaging the outcomes from these 30 runs to derive the final result.

We assess the practical applicability of the proposed MEARO-KELM by evaluating its performance on a score dataset that we collected ourselves. The dataset comprises 20 score features and 120 samples, which are exam scores from 120 students.

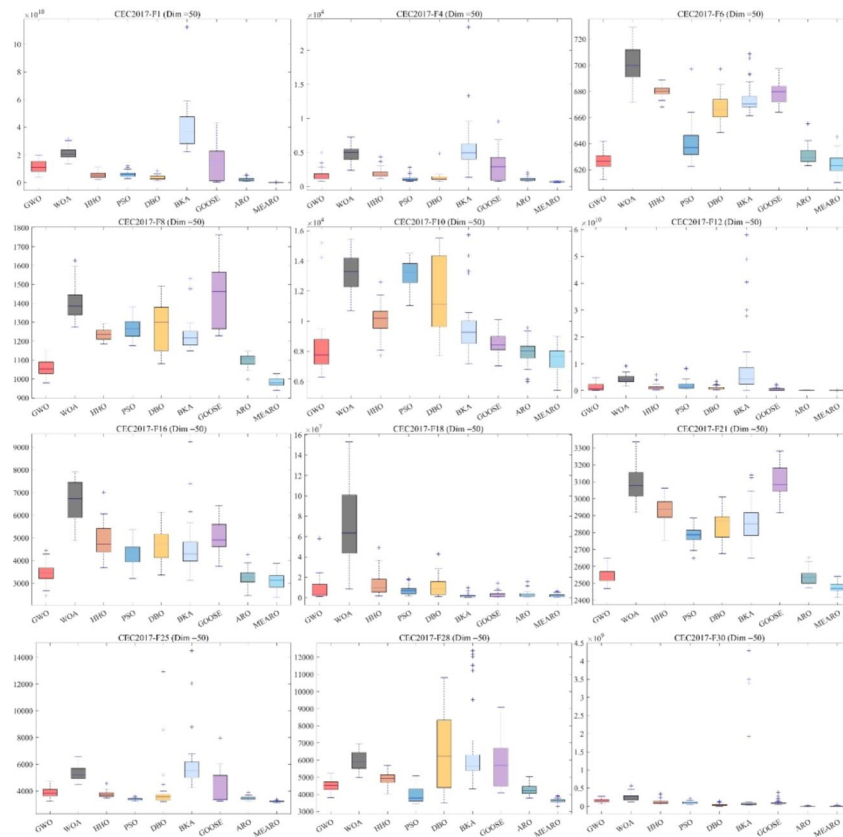


Fig. 9. Boxplot of different algorithms in optimization of the CEC-2017 (Dim = 50).

Classification prediction problem

Performance evaluation metrics

We use standard metrics to gauge the performance of the proposed method in multi-class classification. These metrics encompass accuracy (ACC), Matthews Correlation Coefficient (MCC), sensitivity, and specificity^{71–73}. These metrics are defined based on the confusion matrix as follows:

$$ACC = \frac{1}{n} \sum_{i=1}^n \frac{TP_i + TN_i}{TP_i + FP_i + FN_i + TN_i} \times 100\% \quad (33)$$

$$MCC = \frac{1}{n} \sum_{i=1}^n \frac{TP_i \times TN_i - FP_i \times FN_i}{\sqrt{(TP_i + FP_i) \times (TP_i + FN_i) \times (TN_i + FP_i) \times (TN_i + FN_i)}} \times 100\% \quad (34)$$

$$Sensitivity = \frac{1}{n} \sum_{i=1}^n \frac{TP_i}{TP_i + FN_i} \times 100\% \quad (35)$$

$$Specificity = \frac{1}{n} \sum_{i=1}^n \frac{TN_i}{TN_i + FP_i} \times 100\% \quad (36)$$

Where TP_i denotes the true positives, which are the cases correctly identified as “positive”. TN_i stands for true negatives, representing the cases correctly identified as “negative”. FP_i is the count of false positives, or negative cases mistakenly classified as “positive”. Similarly, FN_i is the count of false negatives, or positive cases wrongly classified as “negative”. MCC is a widely recognized metric that quantifies the correlation coefficient considering both observed and predicted values. A high MCC value suggests that the proposed model is suitable for handling imbalanced datasets. Sensitivity measures the classifier’s ability to detect positives effectively, whereas specificity does the same for negatives.

Different features affect the analysis

To delve deeper into the influence of various factors on the final grades of the tourism service communication courses course, we quantified the relationships between features and outcomes by calculating correlation coefficients. These relationships were then visualized as bar charts for a clearer presentation. As illustrated in Fig. 16, the chart depicts the significance of twenty different attributes on the course’s final grades, using a color gradient that transitions from blue to red to indicate decreasing levels of impact.

The analysis reveals that the top three determinants of academic performance are the interest in learning the course, the frequency of classroom discussions, and the channels used to obtain additional knowledge and information related to the course. In stark contrast, the three factors with the least influence on academic success

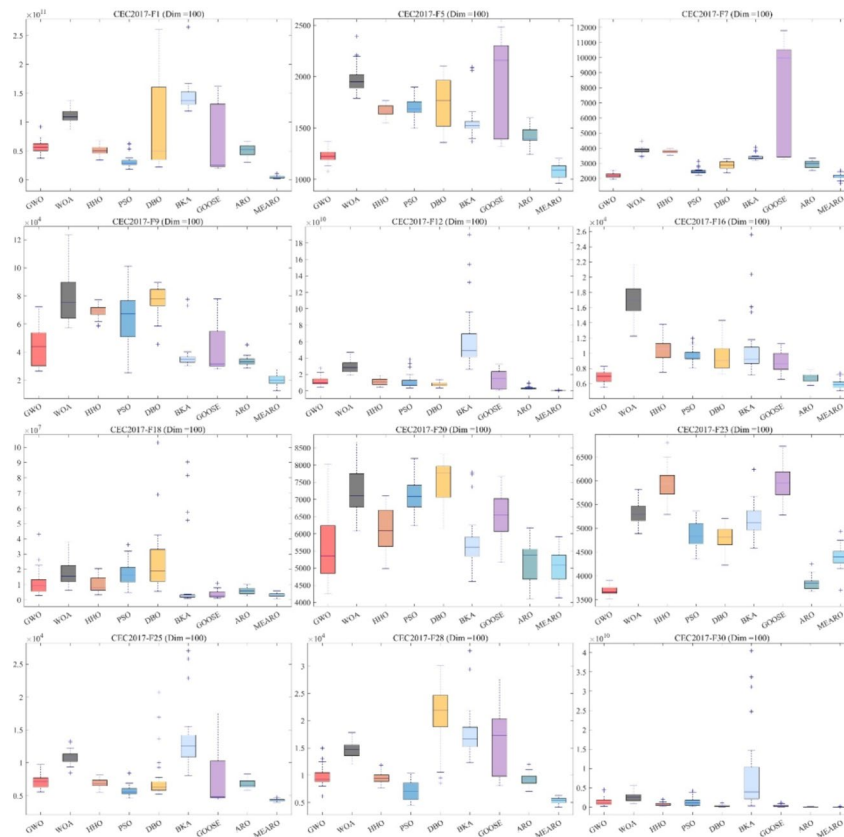


Fig. 10. Boxplot of different algorithms in optimization of the CEC-2017 (Dim = 100).

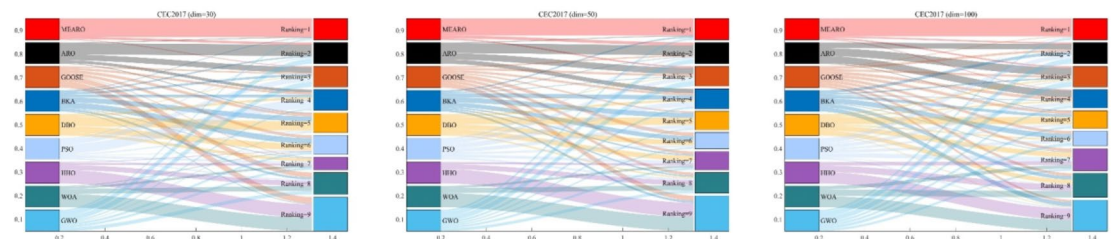


Fig. 11. MEARO and other comparison algorithms on CEC 2017 ranking Sankey chart.

were found to be the methods of reviewing the course, the hometown of the students, and previous learning or practical experience related to the course.

This highlights the paramount importance of engagement and active participation in academic success. Intrinsic motivation and interactive discussions significantly boost performance, while conventional methods like revision techniques and past experiences have a lesser impact. These findings champion proactive learning and keen interest as key to excellence in Tourism Service Communication studies.

Experimental results and analysis

Figures 17 and 18 presents the violin plots comparing the detailed results of GWO-KELM, WOA-KELM, HHO-KELM, PSO-KELM, DBO-KELM, BKA-KELM, GOOSE-KELM, ARO-KELM, and MEARO-KELM on the student performance dataset.

From Fig. 17, it can be seen that the fitness values of the objective function, including the maximum(Max), minimum(Min), median, average(Mean) and standard deviation(STD), all outperform the original ARO and other comparison algorithms. Furthermore, the convergence curves indicate that other comparison algorithms plateau at a minimum error of around 0.0165 and are unable to find better solutions. In contrast, the MEARO presented in this paper continues to seek solutions, achieving an error of approximately 0.013. This demonstrates the effectiveness of the improvement strategies employed in this study, enhancing the optimization performance of the algorithm.

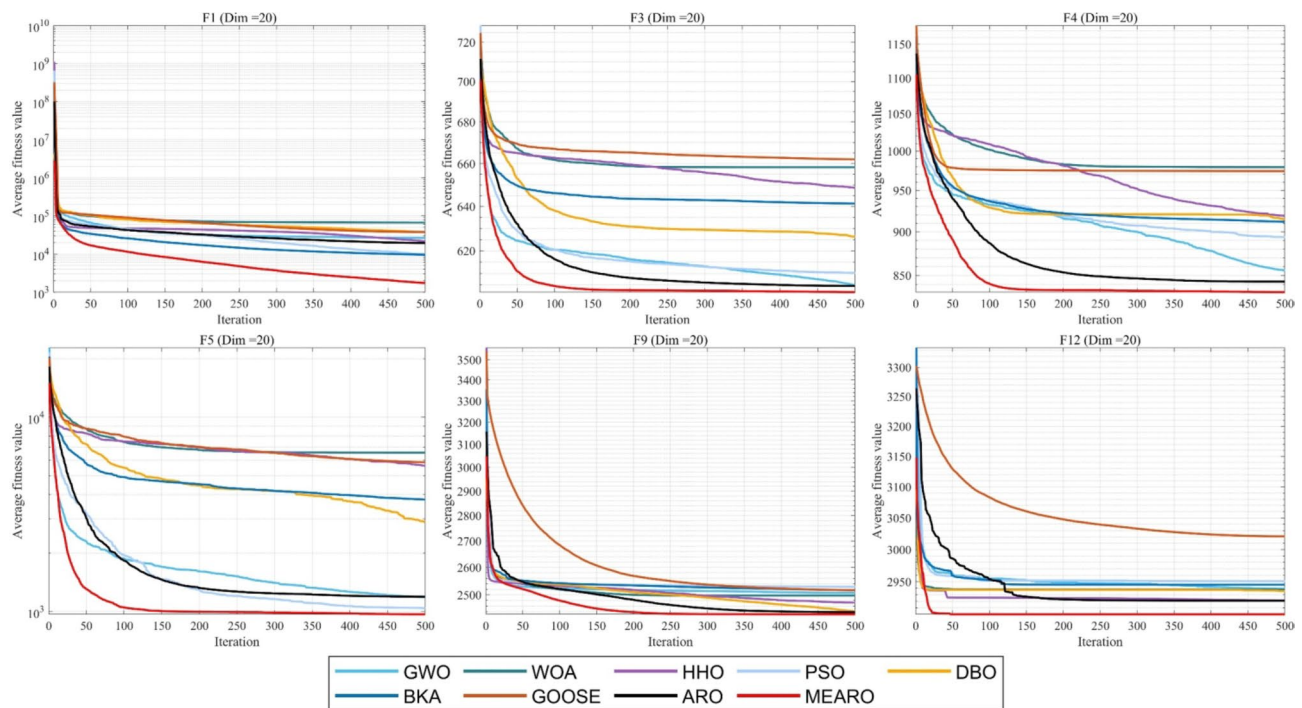


Fig. 12. CEC2022 experimental convergence curve (Dim = 20).

Figure 18 clearly shows that MEARO-KELM outperforms other methods in terms of MCC, sensitivity, and specificity. While its accuracy (ACC) is slightly lower than that of ARO-KELM, MEARO-KELM excels in the other metrics. ARO-KELM's performance, excluding ACC, consistently falls within the lower three rankings for the four metrics. On the other hand, HHO-KELM records the poorest results across all metrics, indicating that its predictive capability for student scores might be less effective compared to the other methods. Furthermore, when considering ACC, MCC, and sensitivity, ARO-KELM produces average values that surpass those of the other models, closely trailing MEARO-KELM. However, the average values generated by ARO-KELM in terms of specificity rank lower, indicating its weakness in identifying negative cases. It's also noteworthy that PSO-KELM's performance across all four metrics is inferior to the proposed MEARO-KELM. Even in ACC, MCC, and sensitivity, it lags behind ARO-KELM. In terms of specificity, ARO-KELM's performance is also surpassed by BKA-KELM, DBO-KELM, and GOOSE-KELM.

Figure 18 makes it clear that the MEARO-KELM classifier generally surpasses the other comparison models. This superiority can be credited to the strong optimization capabilities of the MEARO optimizer we introduced. The notable fluctuation in ARO-KELM's results on the test dataset points to its unstable performance. This further underscores the good optimization performance of the proposed MEARO, which is superior to the other comparison algorithms. It also demonstrates the effectiveness of the three improvement strategies we introduced.

Conclusions and future directions

Within the experimental segment, to test the superiority of MEARO in searching for the global optimum of specific functions, we compared MEARO with several popular metaheuristic algorithms and the standard ARO on the CEC2017 benchmark test functions. The statistical outcomes for each optimization challenge validate that the MEARO, enhanced with our three optimization strategies, exhibits swift convergence and outstanding performance overall. Our experiments show that the MEARO notably surpasses alternative approaches in terms of universal optimality, statistical outcomes, and stability. To further gauge MEARO's efficacy in predicting student grades, we applied the KELM model using various intelligent algorithms for comparison. The results from multiple samples consistently show that the KELM approach based on MEARO excels over other advanced grade prediction models across all four key performance metrics: ACC, MCC, sensitivity, and specificity. In this study, we identified the top three factors influencing academic performance: the frequency of classroom discussions, and the channels used to obtain additional knowledge and information related to the course. In contrast, the least impactful elements were the methods of reviewing the course, the hometown of the students, and previous learning or practical experience related to the course. Therefore, it can be concluded that the applicability of MEARO-KELM has been extended, and the MEARO-KELM can be considered a valuable tool for teachers to make corresponding teaching decisions based on student characteristics, assisting teachers in making accurate decisions during the teaching process.

Through experimental observations on the CEC2017 and CEC2022 test sets, it was found that the MEARO algorithm converges rapidly in most cases. However, there is a risk of getting stuck in local optima at times

ID	Metric	GWO	WOA	HHO	PSO	DBO	BKA	GOOSE	ARO	MEARO
F1	Best	1.8938E+04	2.8494E+04	1.5506E+04	5.0509E+03	1.1151E+04	2.5528E+03	7.0607E+03	9.7598E+03	4.7926E+02
	Worst	3.8070E+04	1.1187E+05	2.7390E+04	3.0577E+04	5.6993E+04	4.6717E+04	1.1788E+05	2.9897E+04	5.7690E+03
	Ave	2.6443E+04	6.7029E+04	2.1778E+04	9.9971E+03	3.7885E+04	9.5798E+03	3.7787E+04	1.9525E+04	1.7446E+03
	Std	5.3067E+03	2.4420E+04	3.4423E+03	4.9102E+03	1.0686E+04	7.9319E+03	2.1811E+04	4.4203E+03	1.4594E+03
F2	Best	4.5210E+02	4.8642E+02	4.8421E+02	4.6605E+02	4.8037E+02	4.5861E+02	4.6158E+02	4.0259E+02	4.0313E+02
	Worst	5.8652E+02	1.5159E+03	8.3206E+02	7.2206E+02	7.0083E+02	2.8153E+03	8.1771E+02	5.5493E+02	5.0364E+02
	Ave	4.9453E+02	7.7531E+02	6.1634E+02	5.2451E+02	5.4124E+02	7.0670E+02	6.2266E+02	4.7524E+02	4.6790E+02
	Std	3.9033E+01	1.8901E+02	1.0879E+02	7.3205E+01	5.2108E+01	5.6691E+02	9.0739E+01	3.4113E+01	1.7248E+01
F3	Best	6.0139E+02	6.3545E+02	6.3993E+02	6.0409E+02	6.0770E+02	6.2717E+02	6.4496E+02	6.0115E+02	6.0003E+02
	Worst	6.1085E+02	7.0587E+02	6.5894E+02	6.2474E+02	6.4787E+02	6.5678E+02	6.8555E+02	6.1058E+02	6.0814E+02
	Ave	6.0492E+02	6.5829E+02	6.4863E+02	6.1025E+02	6.2630E+02	6.4139E+02	6.6204E+02	6.0460E+02	6.0191E+02
	Std	2.1683E+00	1.4662E+01	5.0579E+00	3.9740E+00	1.0617E+01	6.9719E+00	1.2465E+01	2.6708E+00	2.3350E+00
F4	Best	8.1653E+02	8.9051E+02	8.7709E+02	8.5712E+02	8.5220E+02	8.6595E+02	9.2537E+02	8.1298E+02	8.1650E+02
	Worst	9.3322E+02	1.0719E+03	9.5968E+02	9.2747E+02	9.7167E+02	9.5850E+02	1.0487E+03	8.7964E+02	8.7064E+02
	Ave	8.5581E+02	9.7927E+02	9.1866E+02	8.9368E+02	9.1417E+02	9.1187E+02	9.7419E+02	8.4321E+02	8.3167E+02
	Std	2.1524E+01	3.8859E+01	2.0833E+01	1.8966E+01	4.0262E+01	2.2862E+01	3.1903E+01	1.5478E+01	1.1729E+01
F5	Best	9.2857E+02	3.5159E+03	3.4260E+03	9.3750E+02	1.2309E+03	2.3608E+03	3.0777E+03	9.1145E+02	9.0050E+02
	Worst	2.4446E+03	9.2095E+03	9.2231E+03	1.3253E+03	5.2918E+03	6.6274E+03	8.6575E+03	2.1702E+03	1.2241E+03
	Ave	1.1847E+03	6.5666E+03	5.6285E+03	1.0407E+03	2.8964E+03	3.7692E+03	5.8637E+03	1.1923E+03	9.6788E+02
	Std	2.9472E+02	1.5397E+03	1.4438E+03	9.9118E+01	9.5561E+02	9.7208E+02	1.4886E+03	2.8040E+02	7.7223E+01
F6	Best	7.1706E+03	4.7163E+04	4.8073E+04	2.0894E+05	2.2150E+03	3.2279E+03	2.4519E+03	1.8685E+03	1.9036E+03
	Worst	8.9548E+07	1.9276E+07	6.2724E+05	3.0507E+08	1.0669E+07	8.7023E+08	2.4761E+04	1.7766E+04	2.0722E+04
	Ave	3.0071E+06	2.8541E+06	2.9973E+05	1.1812E+07	3.6902E+05	5.0987E+07	8.4705E+03	6.7072E+03	7.1570E+03
	Std	1.6345E+07	4.7870E+06	1.7313E+05	5.5404E+07	1.9453E+06	1.8005E+08	6.9582E+03	4.7577E+03	4.4767E+03
F7	Best	2.0309E+03	2.1110E+03	2.0923E+03	2.0387E+03	2.0471E+03	2.0700E+03	2.1190E+03	2.0098E+03	2.0163E+03
	Worst	2.1340E+03	2.3138E+03	2.2000E+03	2.1611E+03	2.2430E+03	2.2175E+03	2.5104E+03	2.0738E+03	2.1237E+03
	Ave	2.0754E+03	2.1929E+03	2.1336E+03	2.0752E+03	2.1280E+03	2.1113E+03	2.2839E+03	2.0408E+03	2.0361E+03
	Std	2.5377E+01	5.3470E+01	2.7467E+01	2.7515E+01	4.8212E+01	2.9055E+01	1.1518E+02	1.4183E+01	2.0268E+01
F8	Best	2.2241E+03	2.2374E+03	2.2423E+03	2.2174E+03	2.2234E+03	2.2293E+03	2.2489E+03	2.2015E+03	2.2007E+03
	Worst	2.4664E+03	2.4059E+03	2.5907E+03	2.4743E+03	2.4127E+03	2.4693E+03	2.9622E+03	2.3230E+03	2.2232E+03
	Ave	2.2634E+03	2.3161E+03	2.3958E+03	2.2997E+03	2.2991E+03	2.3187E+03	2.5801E+03	2.2191E+03	2.2127E+03
	Std	6.8938E+01	5.9445E+01	8.0832E+01	7.4518E+01	5.4485E+01	6.1445E+01	1.7217E+02	2.1481E+01	9.9931E+00
F9	Best	2.4068E+03	2.4214E+03	2.4276E+03	2.4205E+03	2.4000E+03	2.4206E+03	2.4009E+03	2.4027E+03	2.4000E+03
	Worst	2.5906E+03	2.5500E+03	2.5500E+03	2.6164E+03	2.5500E+03	2.5500E+03	2.8040E+03	2.5503E+03	2.5504E+03
	Ave	2.5071E+03	2.4975E+03	2.4722E+03	2.5287E+03	2.4414E+03	2.5161E+03	2.5164E+03	2.4387E+03	2.4313E+03
	Std	4.4043E+01	4.0553E+01	4.5857E+01	4.1122E+01	5.7868E+01	3.0971E+01	1.0011E+02	5.7191E+01	6.0564E+01
F10	Best	2.5006E+03	2.5063E+03	2.5074E+03	2.5449E+03	2.5000E+03	2.5028E+03	2.5001E+03	2.5003E+03	2.5000E+03
	Worst	4.4909E+03	6.1015E+03	2.5500E+03	5.4872E+03	2.5748E+03	4.6309E+03	6.5202E+03	3.1251E+03	2.8916E+03
	Ave	2.7466E+03	3.2262E+03	2.5178E+03	3.0040E+03	2.5318E+03	2.7353E+03	4.2048E+03	2.5558E+03	2.5425E+03
	Std	5.7121E+02	1.3054E+03	1.6430E+01	1.0116E+03	2.6006E+01	6.1478E+02	1.5904E+03	1.5495E+02	1.0665E+02
F11	Best	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03	2.6000E+03
	Worst	3.2209E+03	3.5084E+03	2.8750E+03	3.3706E+03	3.2883E+03	2.8750E+03	3.5902E+03	2.8750E+03	2.8750E+03
	Ave	2.6777E+03	2.7345E+03	2.7403E+03	2.7111E+03	2.6916E+03	2.7406E+03	2.7732E+03	2.6619E+03	2.6318E+03
	Std	1.5588E+02	2.6843E+02	1.0732E+02	1.7513E+02	1.4255E+02	1.2026E+02	1.7924E+02	1.0428E+02	8.3272E+01
F12	Best	2.9000E+03	2.9000E+03	2.9000E+03	2.9000E+03	2.9000E+03	2.9000E+03	2.9003E+03	2.9001E+03	2.9000E+03
	Worst	3.1787E+03	3.1994E+03	3.0000E+03	3.2559E+03	3.1642E+03	3.6020E+03	3.7330E+03	3.0147E+03	2.9000E+03
	Ave	2.9391E+03	2.9378E+03	2.9211E+03	2.9508E+03	2.9353E+03	2.9454E+03	3.0208E+03	2.9207E+03	2.9000E+03
	Std	8.1783E+01	8.7289E+01	4.0186E+01	9.9022E+01	7.6794E+01	1.3058E+02	2.0780E+02	4.1723E+01	2.1065E-03
Mean		4.58	7.00	6.08	5.50	4.67	5.42	8.00	2.67	1.08
Ranking		3	8	7	6	4	5	9	2	1

Table 7. Experimental evaluation outcomes utilizing CEC202 test functions (dim = 20).

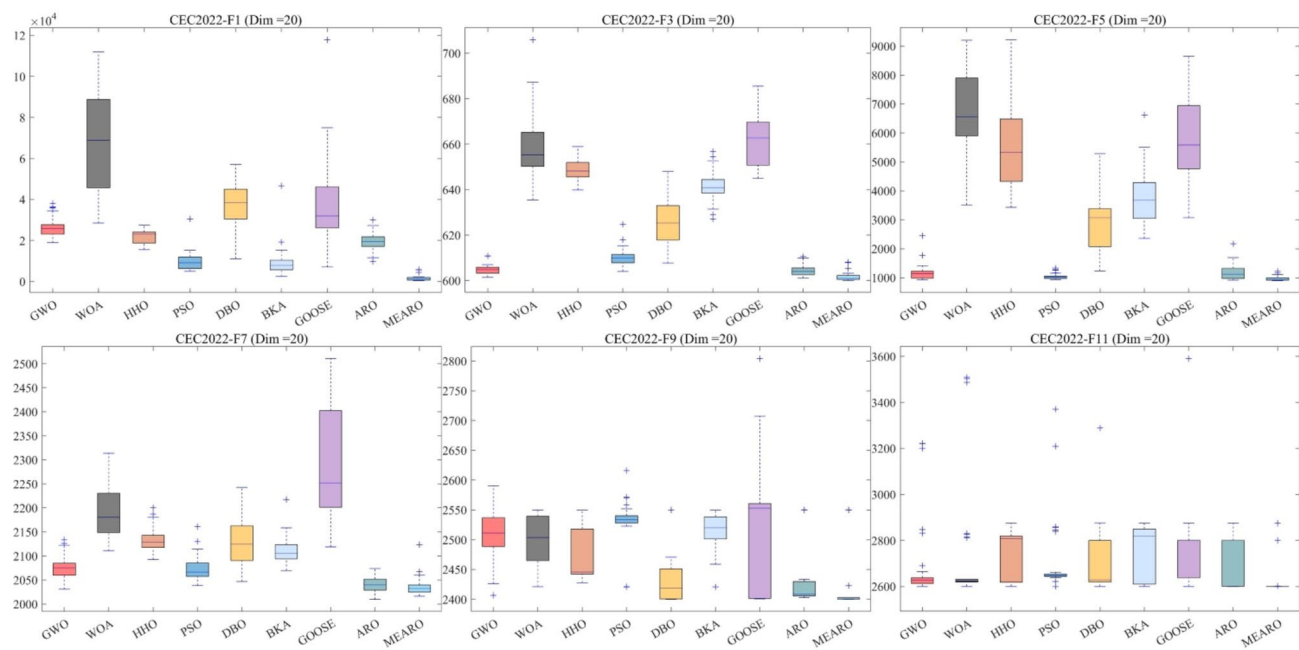


Fig. 13. Boxplot of different algorithms in optimization of the CEC-2022 (Dim = 20).

MEARO VS.		CEC2017 (Dim = 30)	CEC2017 (Dim = 50)	CEC2017 (Dim = 100)	CEC2022 (Dim = 20)
Wilcoxon rank-sum test(+/-/-)	GWO	(26/0/4)	(24/0/6)	(30/0/0)	(12/0/0)
	WOA	(30/0/0)	(30/0/0)	(30/0/0)	(12/0/0)
	HHO	(30/0/0)	(30/0/0)	(30/0/0)	(12/0/0)
	PSO	(26/0/4)	(29/0/1)	(28/0/2)	(12/0/0)
	DBO	(30/0/0)	(30/0/0)	(30/0/0)	(9/0/3)
	BKA	(27/0/3)	(30/0/0)	(26/0/4)	(12/0/0)
	GOOSE	(30/0/0)	(28/0/2)	(27/0/3)	(11/0/1)
	ARO	(24/0/6)	(20/0/10)	(29/0/1)	(9/0/3)
	Overall(+/-/-)	(223/0/17)	(221/0/19)	(230/0/10)	(101/0/7)

Table 8. The wilcoxon rank-sum test conducted on the CEC2017 and CEC2022 dataset.

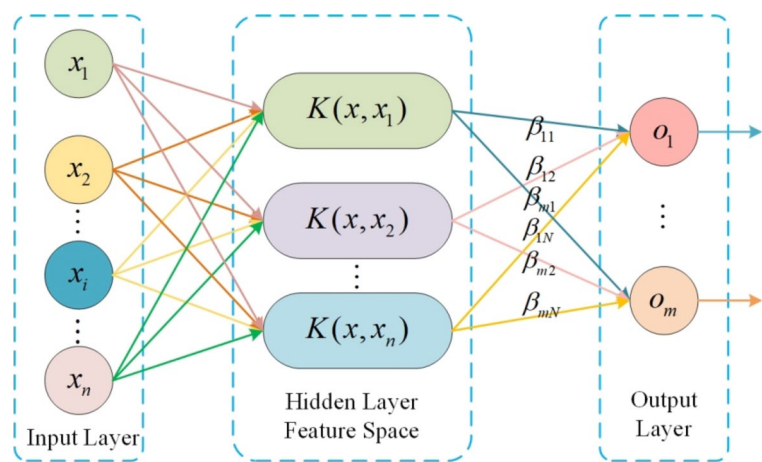


Fig. 14. Structure diagram of KELM model.

Parameter	Search range
Penalty parameter (C)	[1,20]
Kernel parameter (c)	[1,20]

Table 9. Optimized parameters and their ranges.

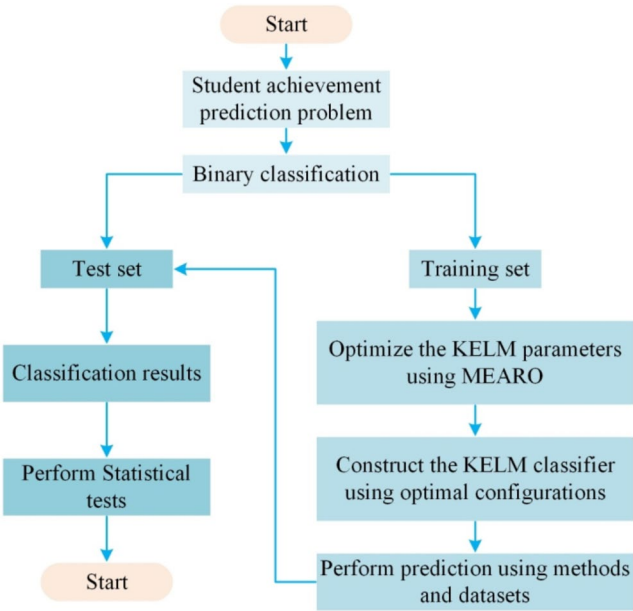


Fig. 15. The general framework of the prediction model utilizing MEARO-KELM.

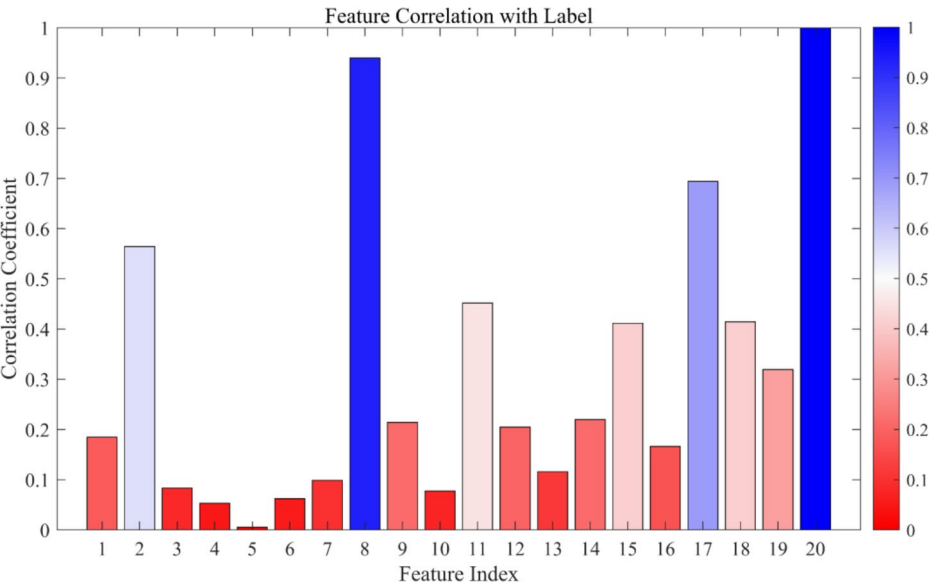


Fig. 16. Correlation coefficient diagram of feature and label.

(e.g., CEC2017-F14, F18, F22, F22, F26 and CEC2022-F6), and the smoothness of the optimization process with MEARO cannot be guaranteed. Moreover, there is a possibility of stagnation due to insufficient population diversity. Other issues require further testing for validation.

In future research, integrating MEARO with other algorithms for collaborative optimization can fully leverage leveraging the unique advantages of each algorithm to further improve the accuracy and robustness

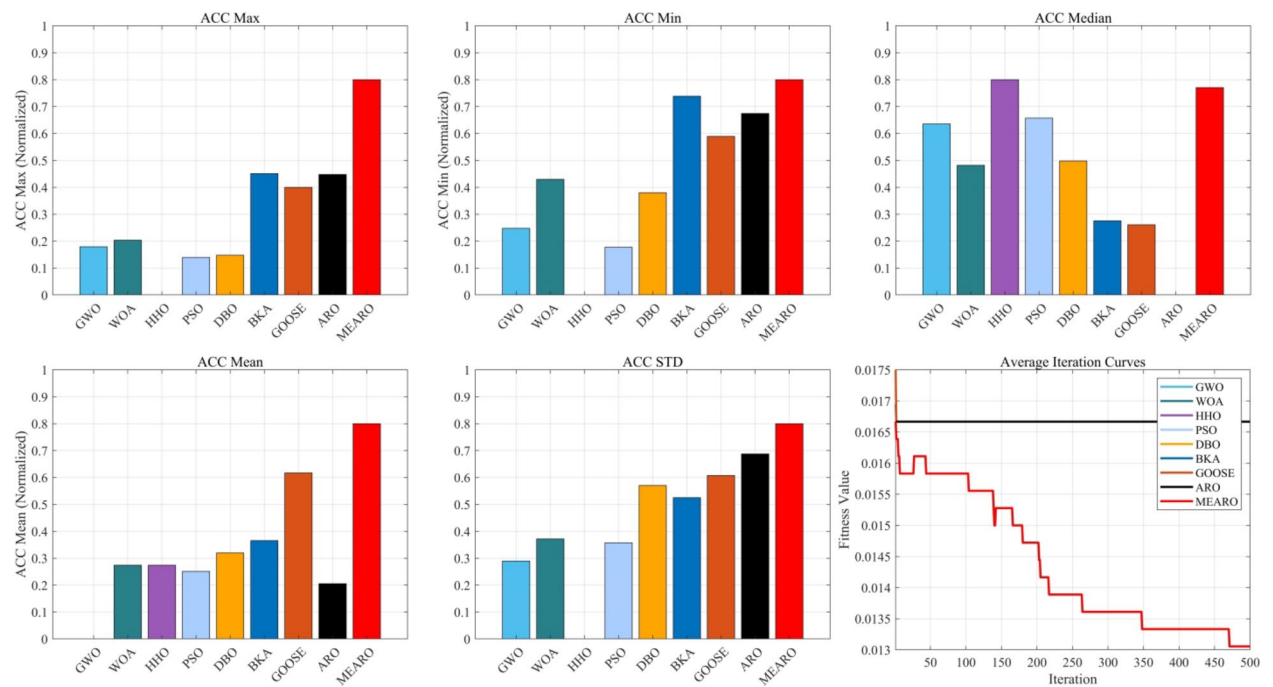


Fig. 17. Fitness value and convergence curve of objective function.

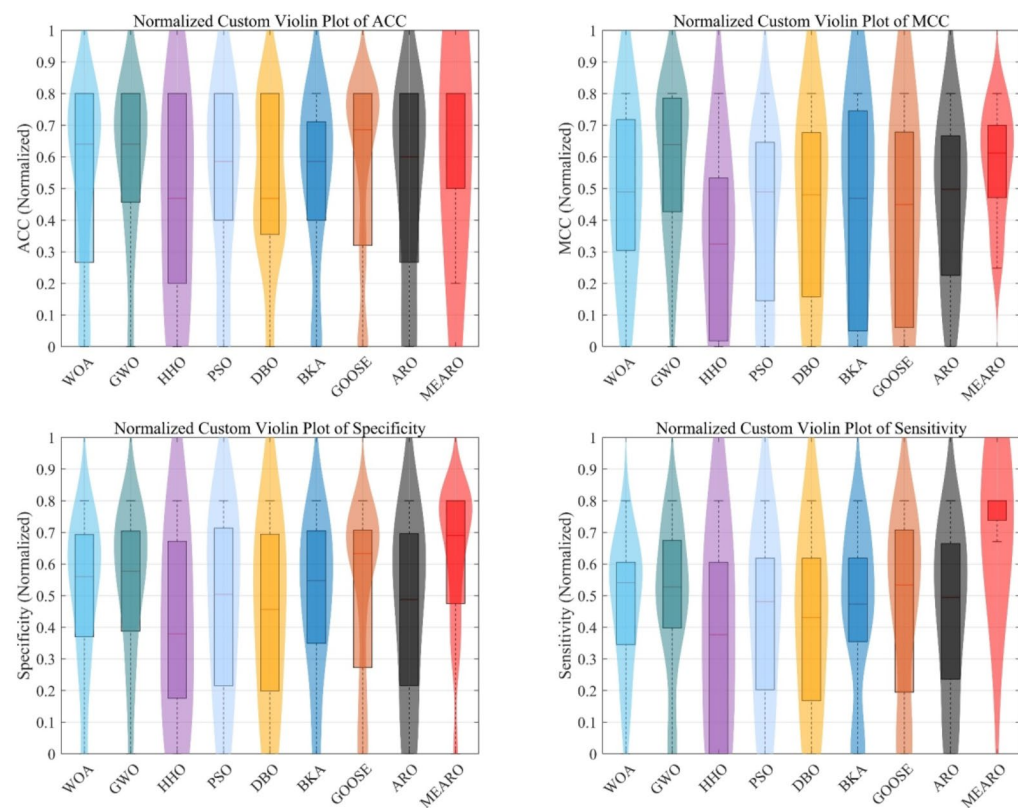


Fig. 18. Violin plot of MEARO-KELM and competitors in terms of ACC, MCC, sensitivity, and specificity on dataset.

of problem-solving. Additionally, introducing artificial intelligence and neural networks techniques to improve MEARO's adaptability and self-learning capabilities can help it accommodate variations across diverse problem domains and contextual settings. This would allow it to glean insights from data and adapt its parameters and strategies accordingly, improving the effectiveness and adaptability of the solutions. The MEARO-KELM model we introduced can be adapted to tackle various real-world classification challenges. Integrating this model with feature selection techniques can streamline computational processes and boost classification accuracy. Additionally, we intend to employ different versions of our MEARO optimization core to tackle more intricate optimization problems, particularly in the realm of optimal resource allocation. By continuously evolving and expanding the capabilities of MEARO and its applications, we aim to provide more powerful and versatile tools for solving various optimization and classification problems in both academic and practical settings. In addition, MEARO can be extended to other educational datasets or explored in other fields, such as healthcare or finance.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

Zhuyin Jia wrote the original draft and performed review & editing, visualization, formal analysis, methodology,

and funding acquisition. Xiaodan Qu handled software development, review & editing, visualization, formal analysis, methodology, and validation. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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